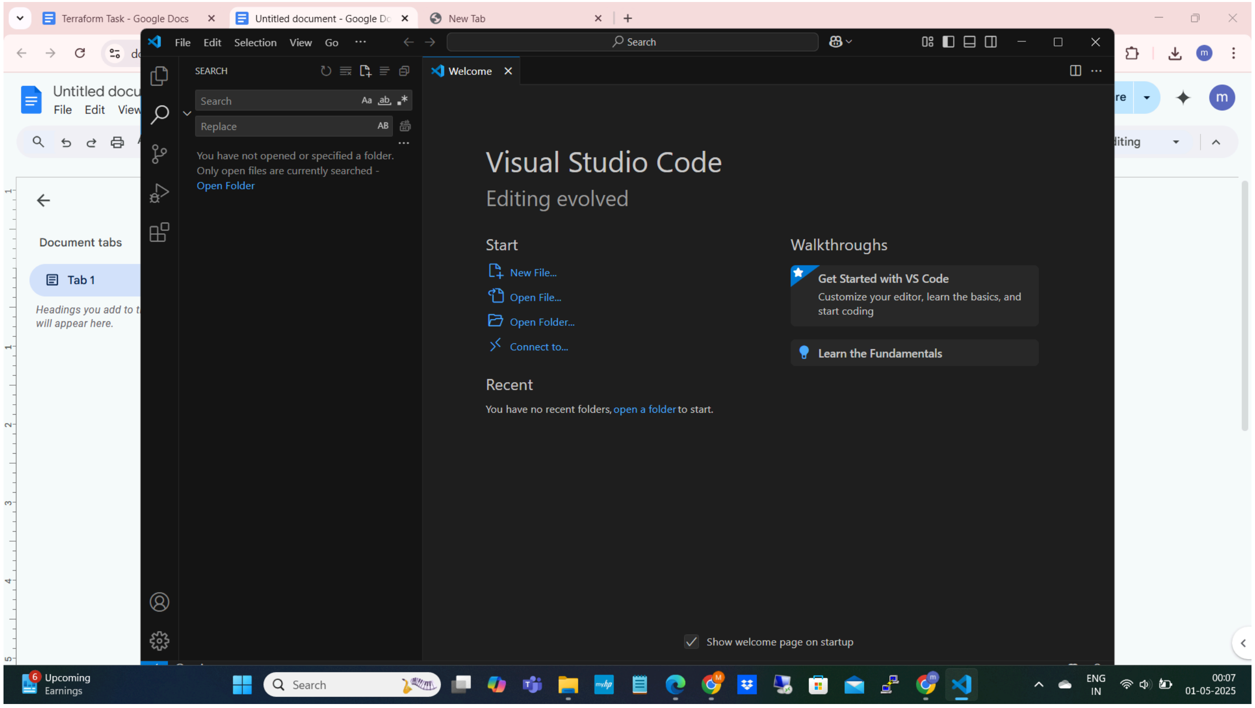
Terraform Task

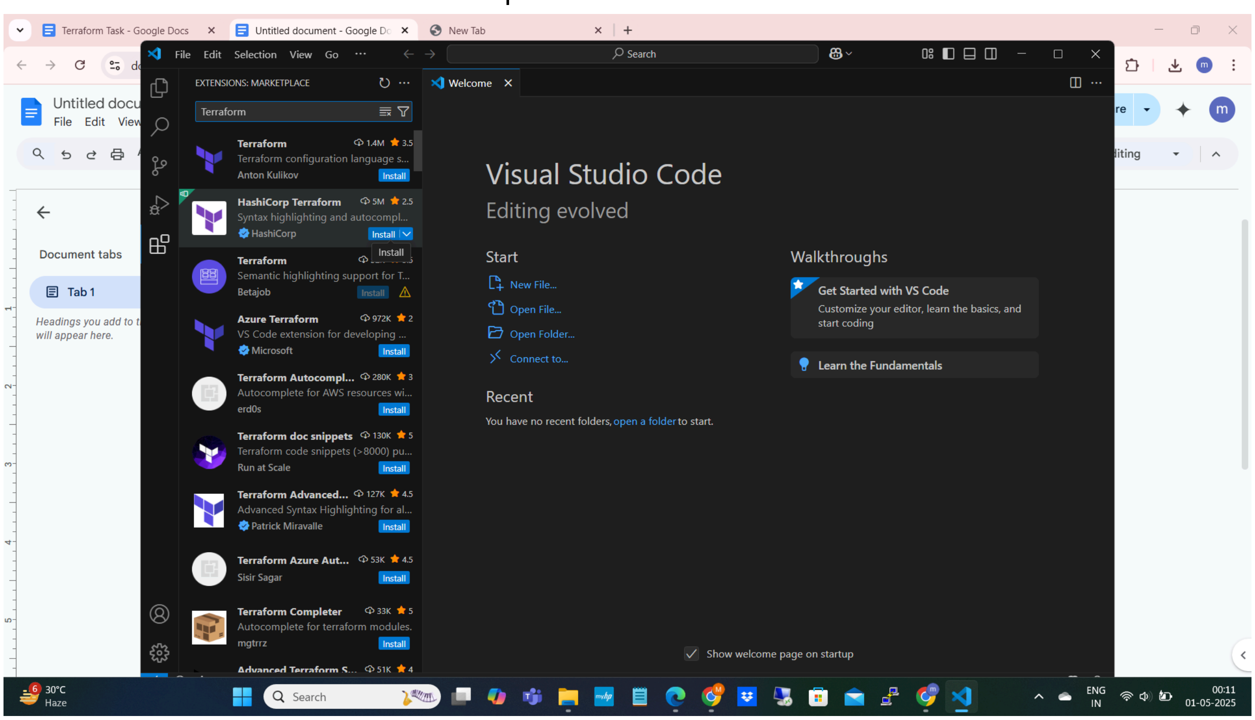
Installed Visual Studio

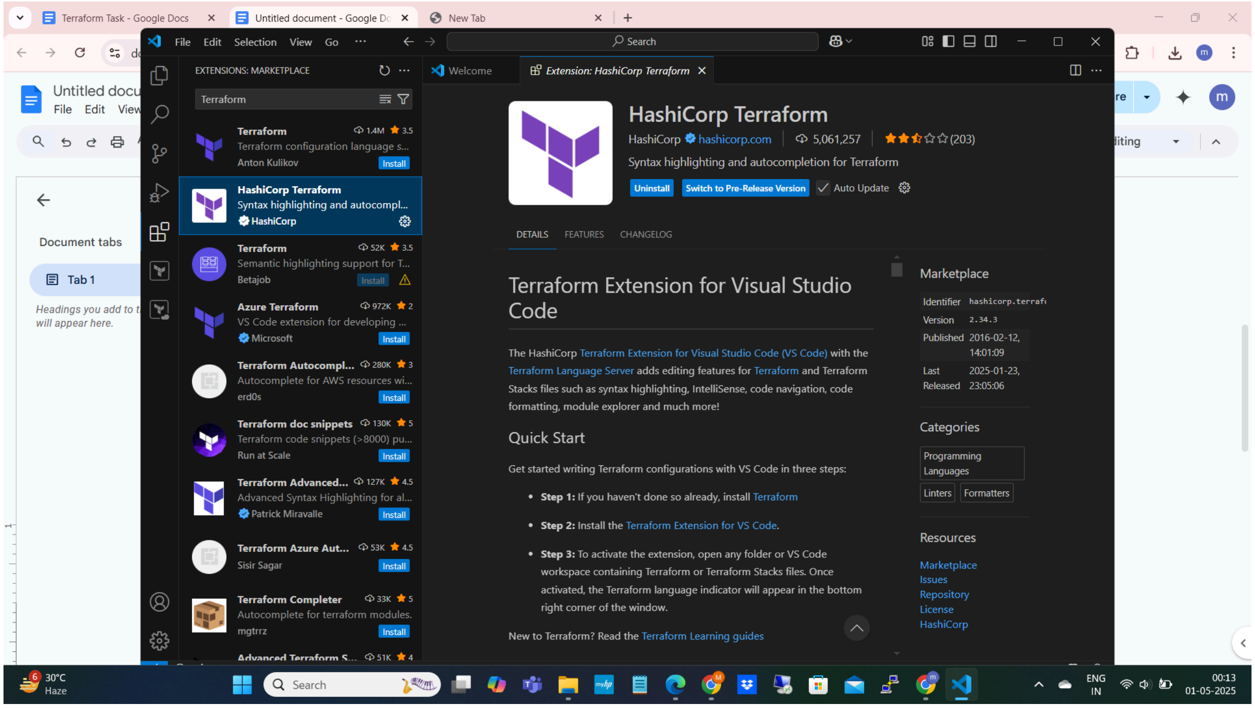


Install Terraform Extension:

Go to Extensions tab and Search for the Terraform

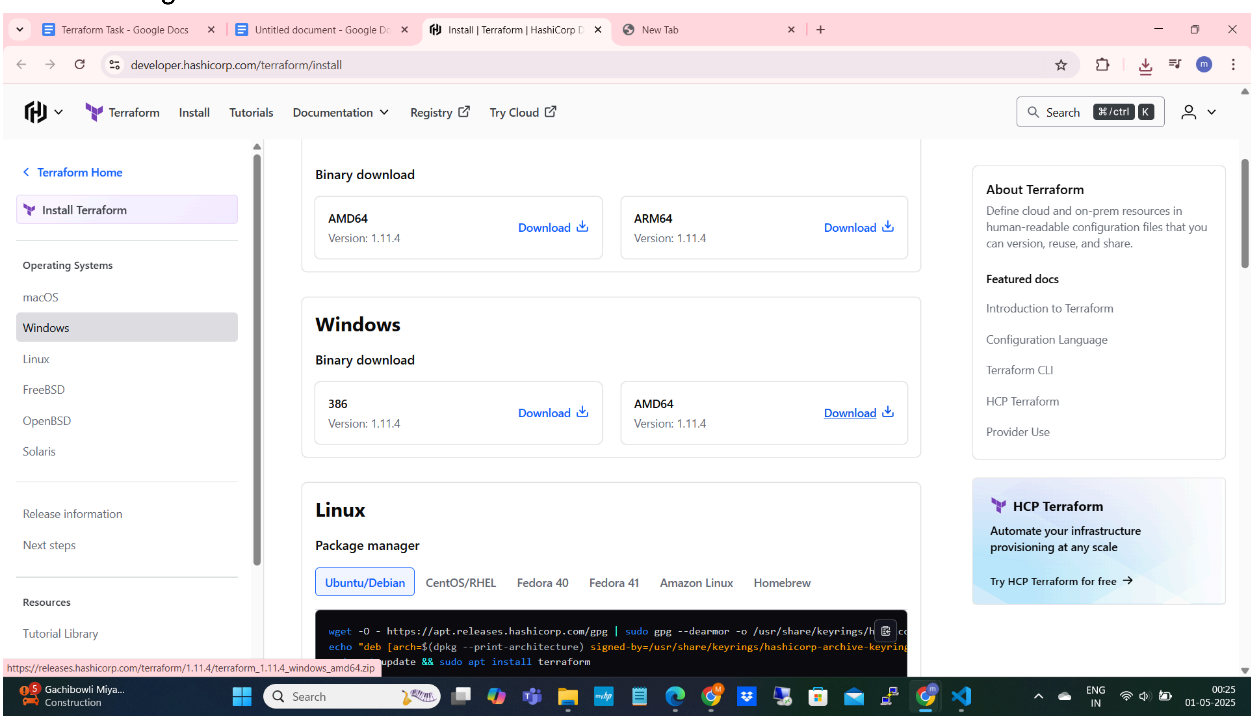
Install the extension from the HashiCorp



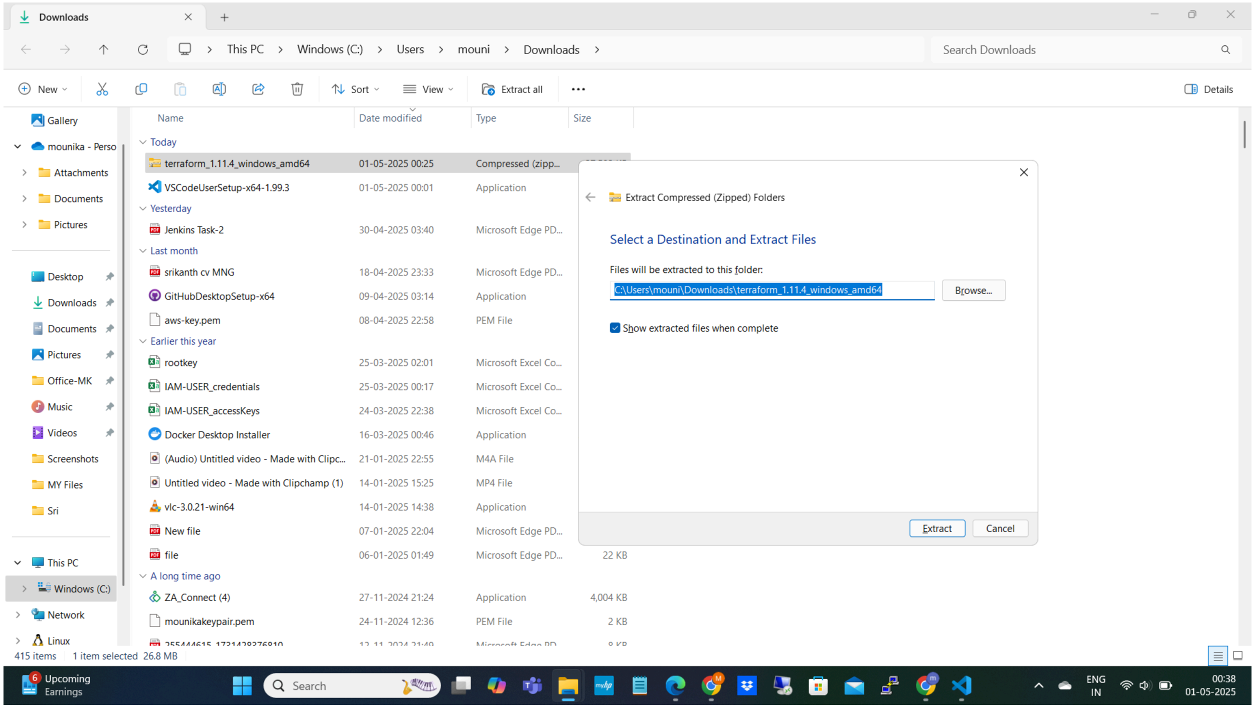


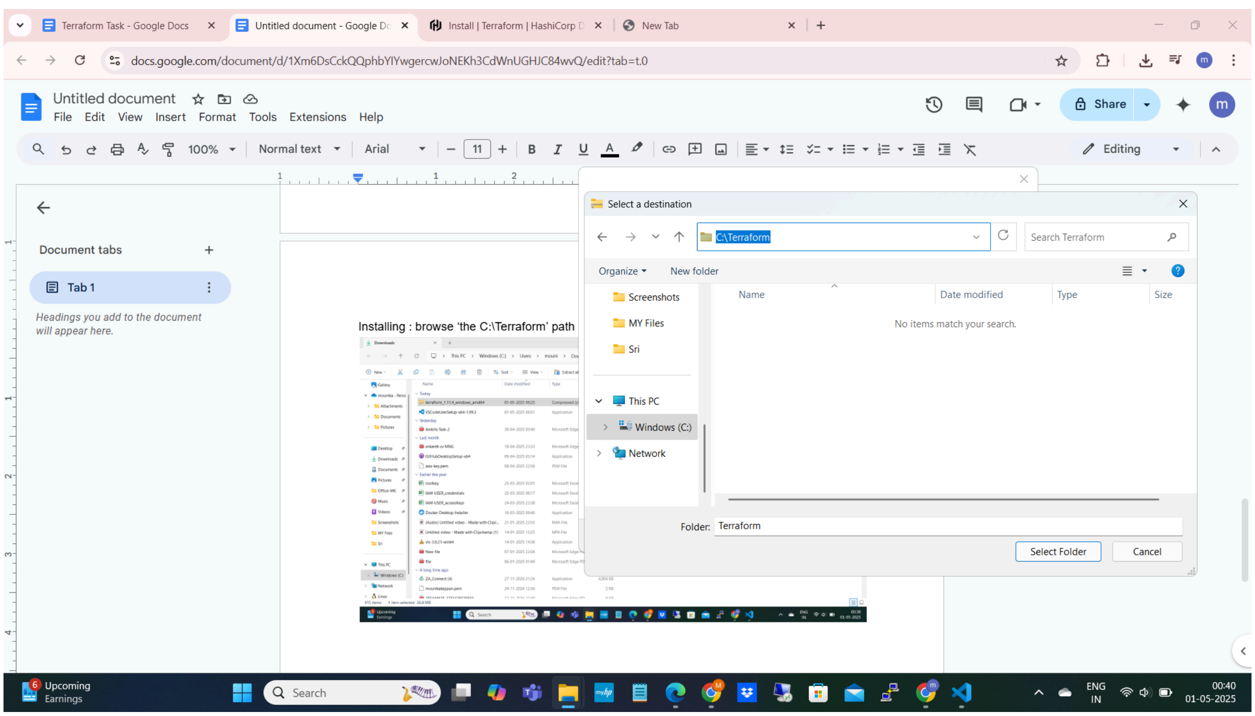
Installing Terraform in My Local machine:

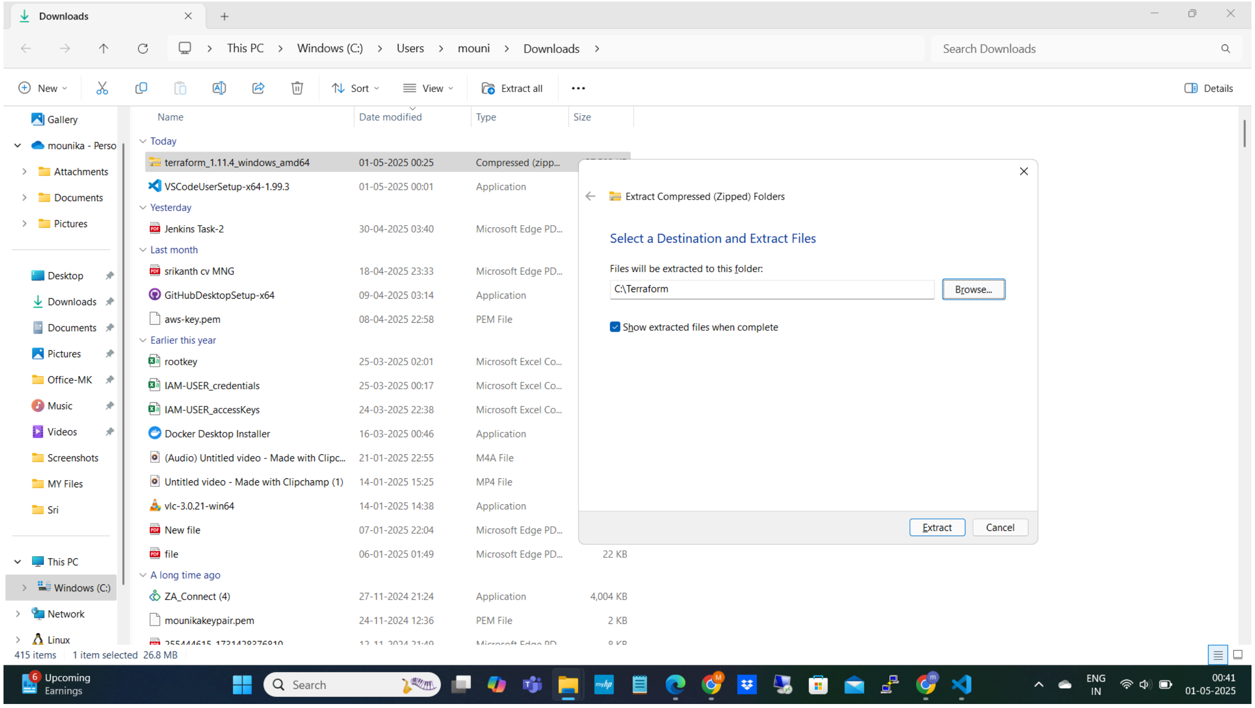
Down loading the Terraform.exe file



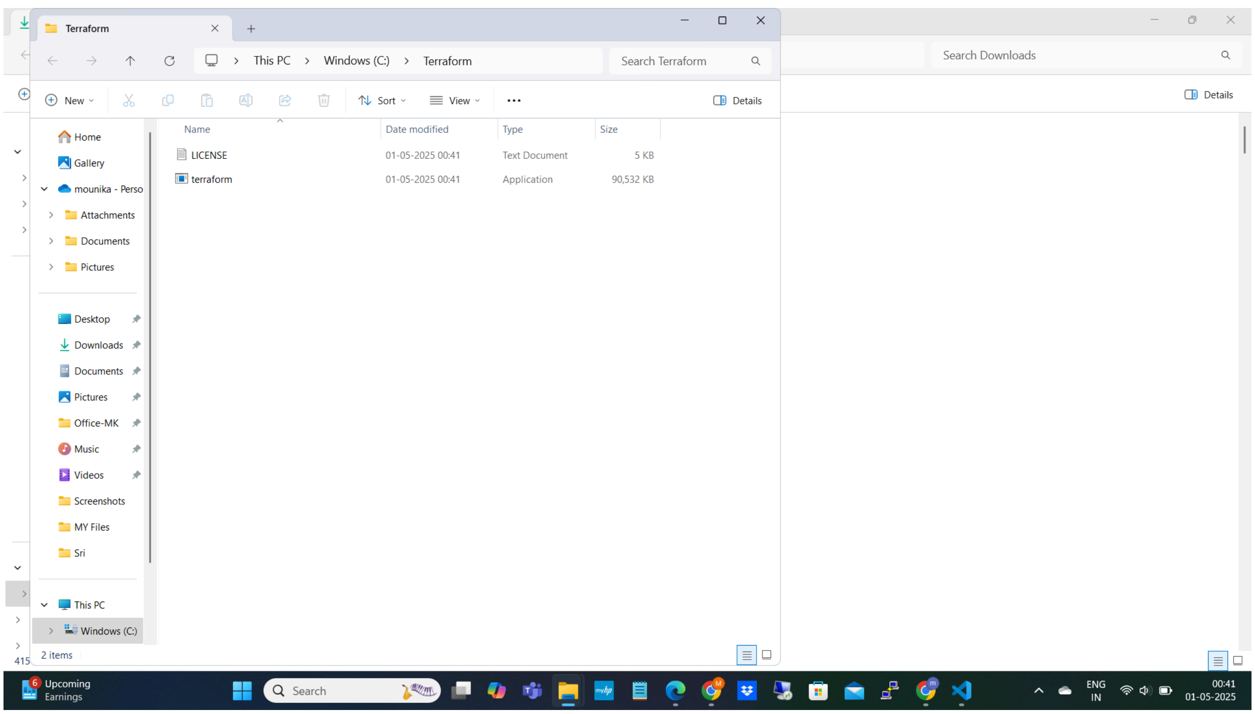
Installing: browse ‘the C:\Terraform’ path





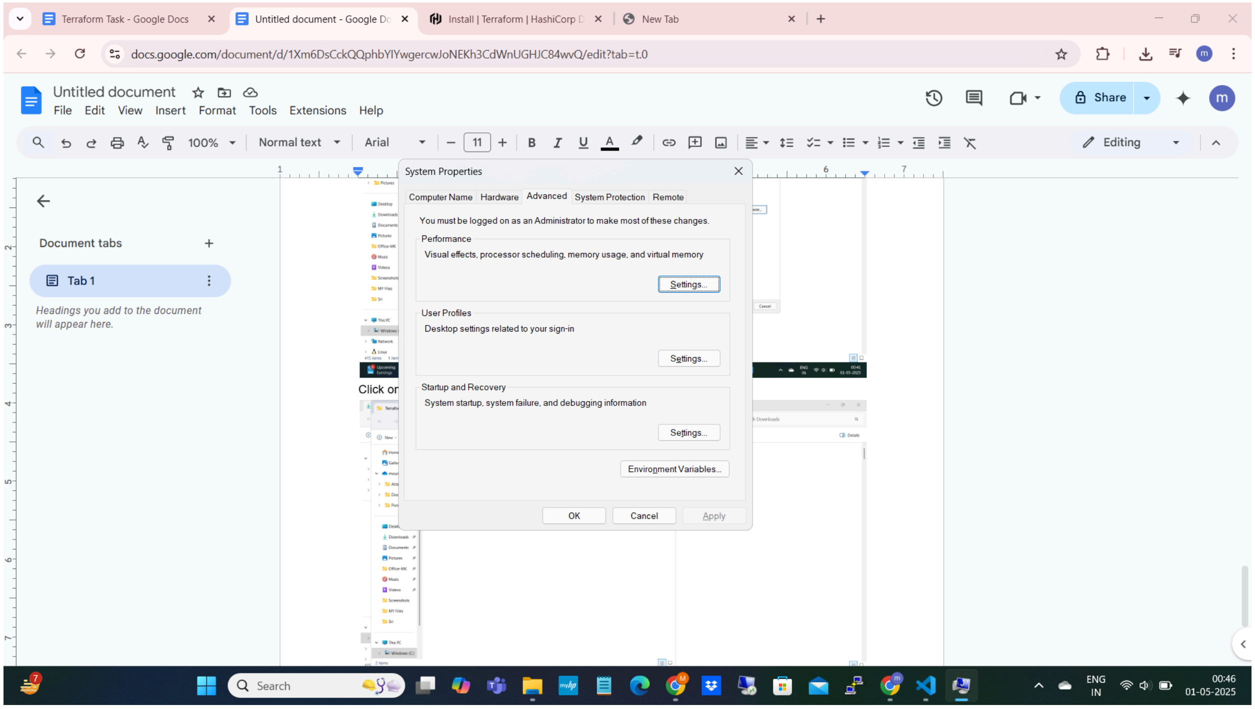


Click on Extract

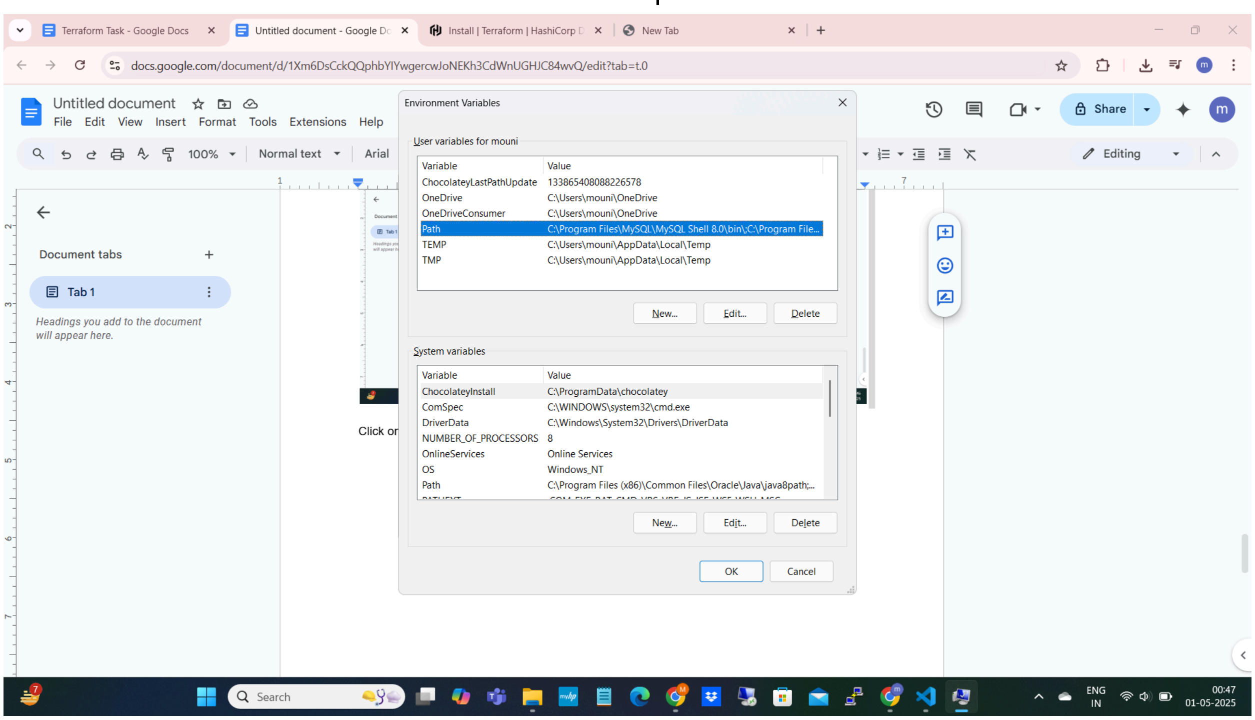


Now Set Environment Variable

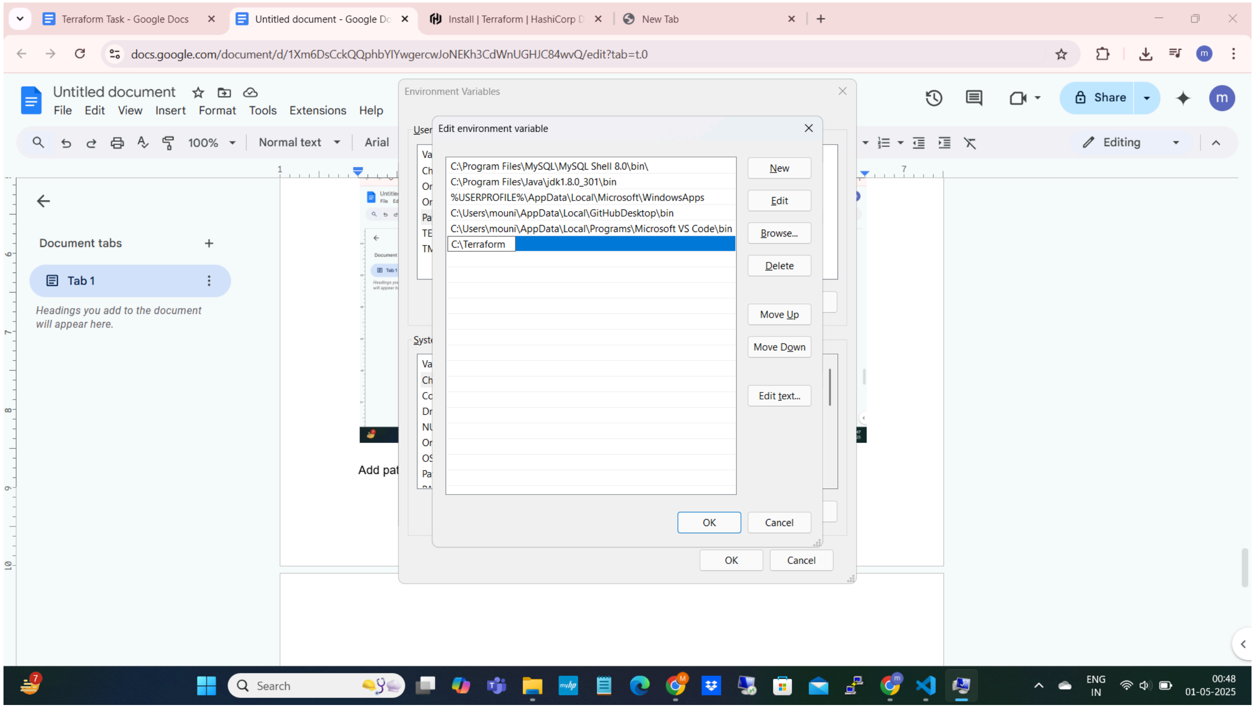
Advanced system settings:



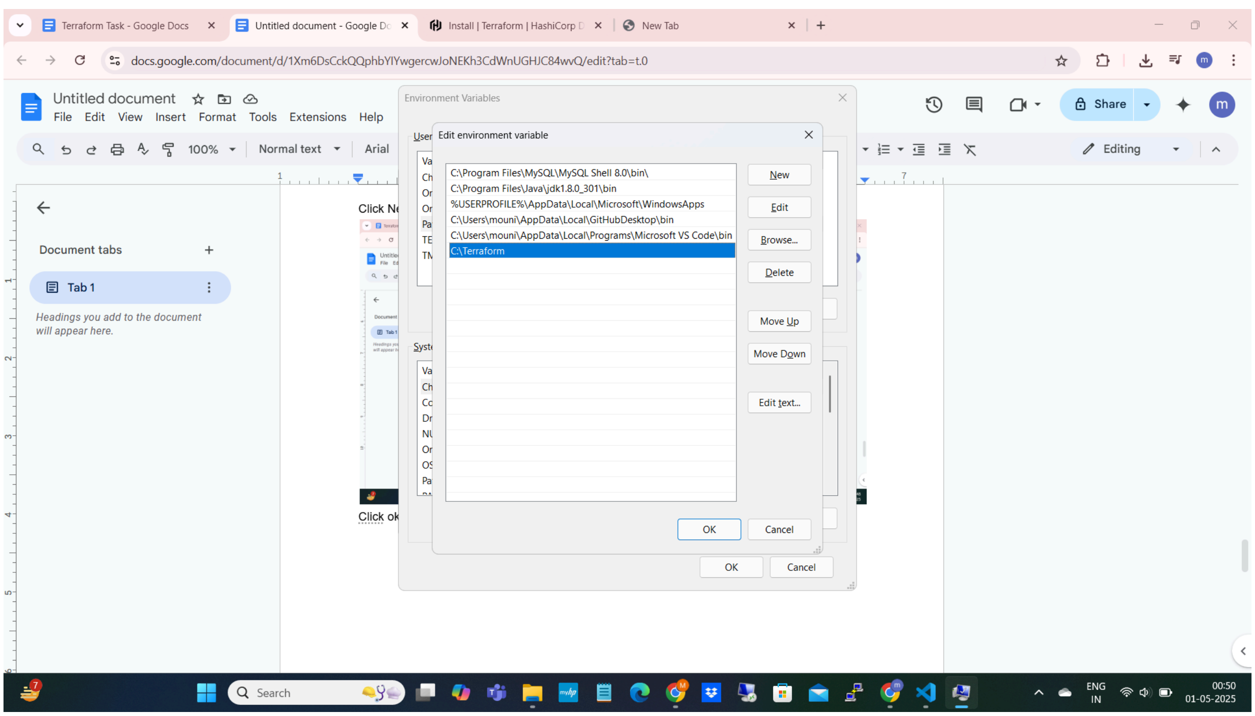
Click on Environment variables and Select the the path variable



Click New and Add the path **C:\Terraform**

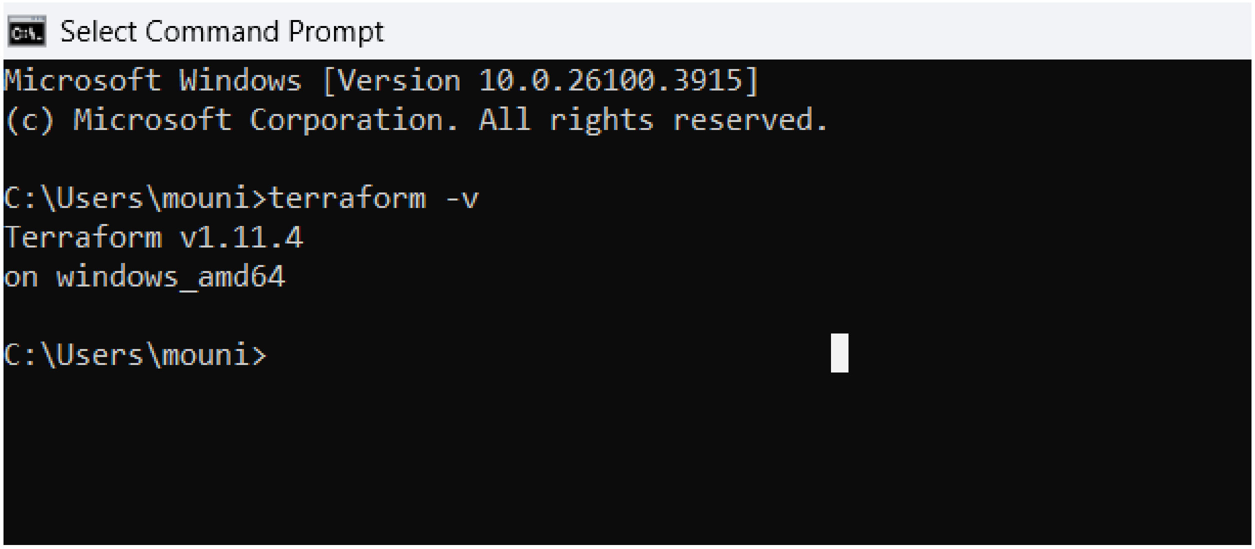


Click ok

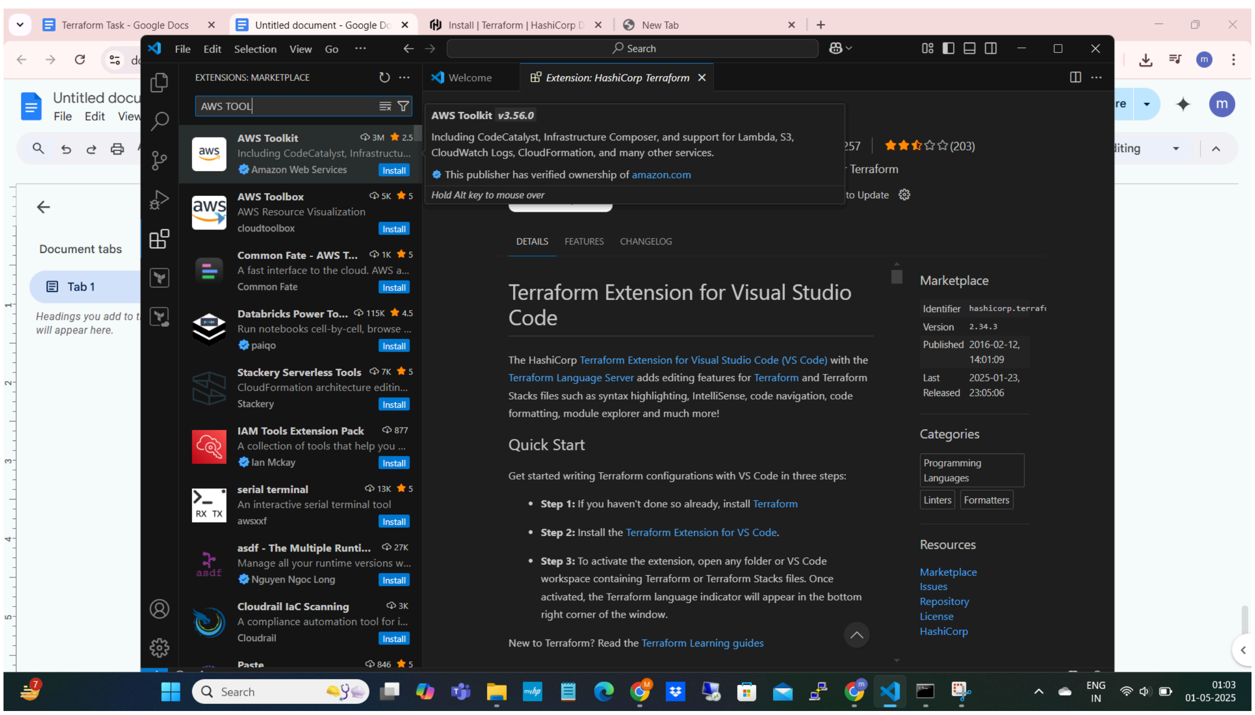


Then click ok

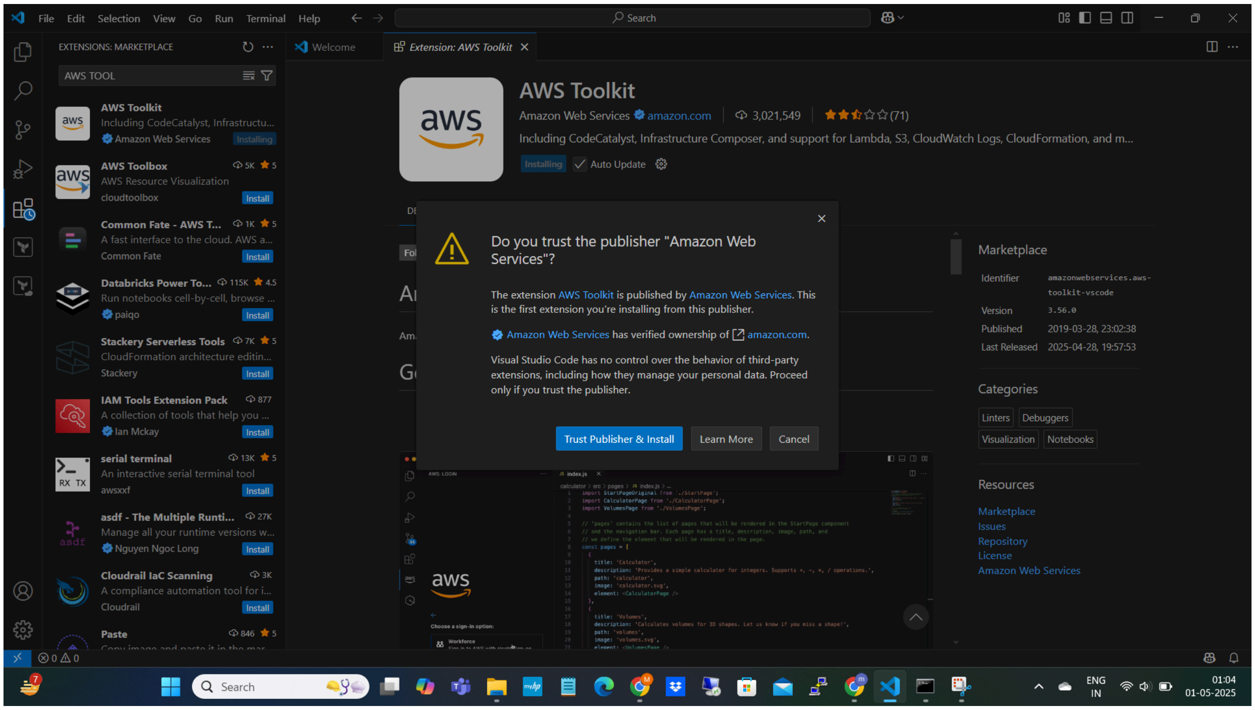
Verify installation
$ terraform -v



Configure AWS Toolkit for VS Code:
Search for AWS toolkit in visual studio and select it



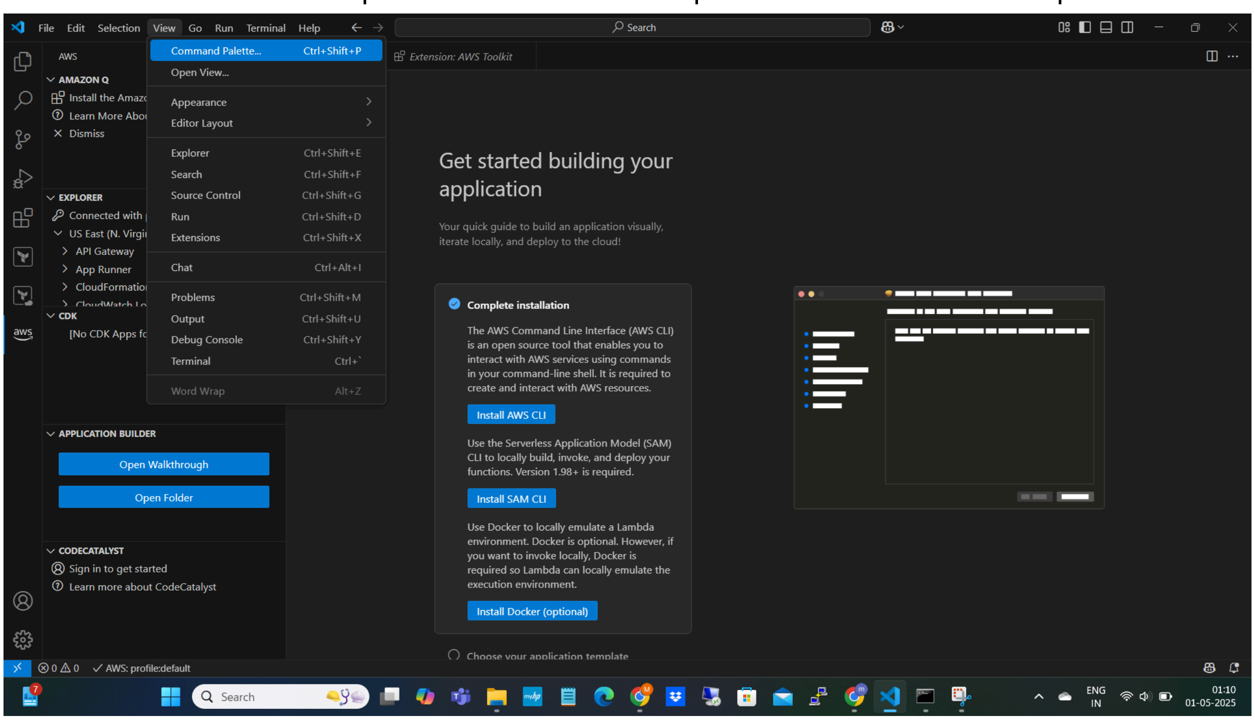
Click on Trust Publisher & install



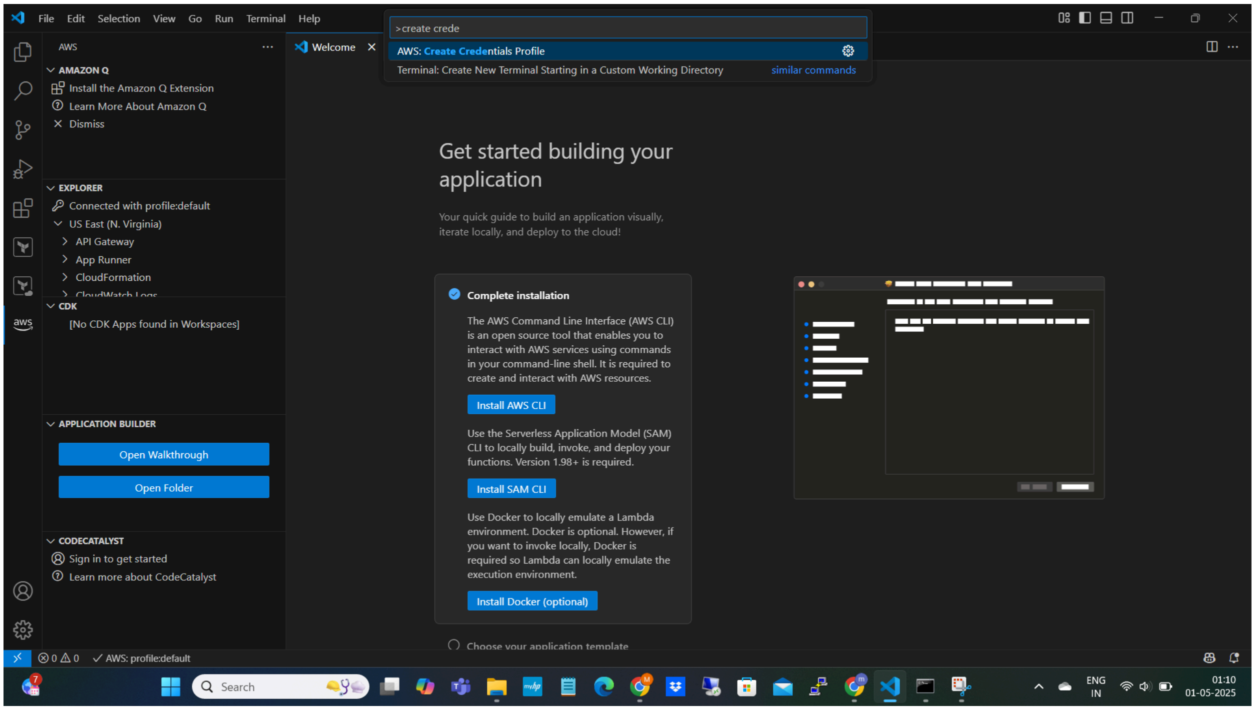
Installation done

Now Configure the AWS account credentials

Click on AWS icon from left panel →View–Command palette→Create credentials profile



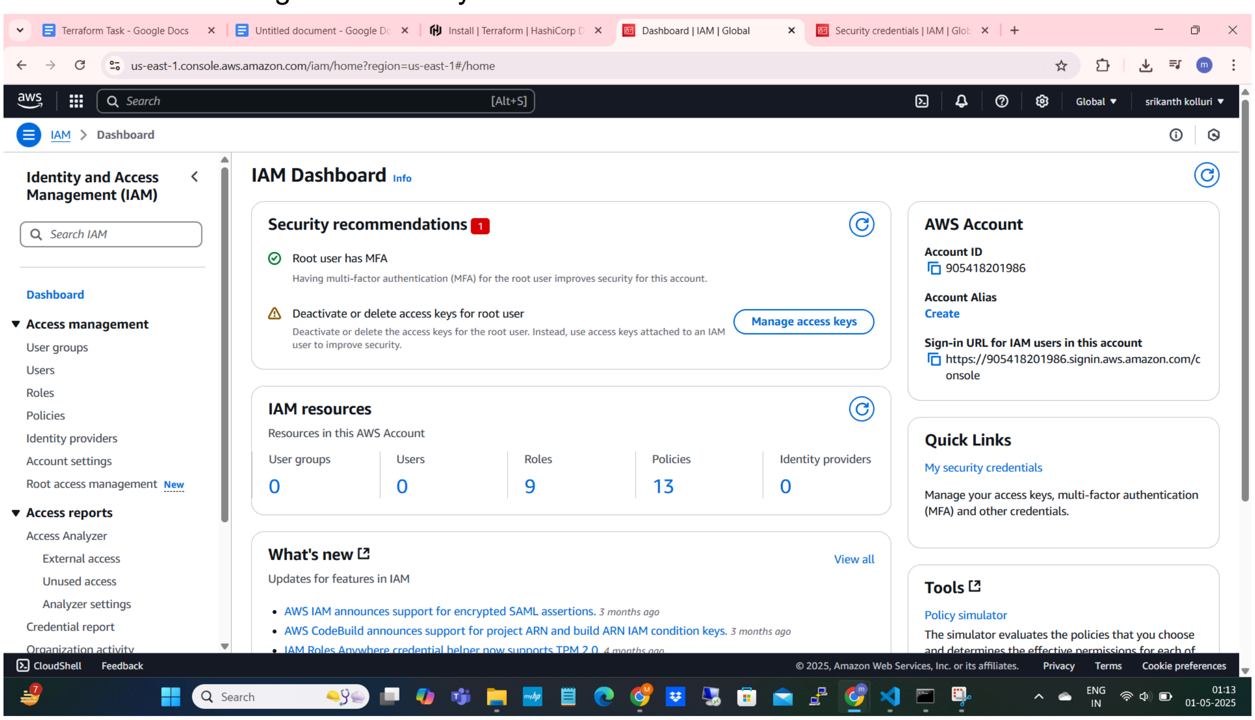
Create credentials Profile

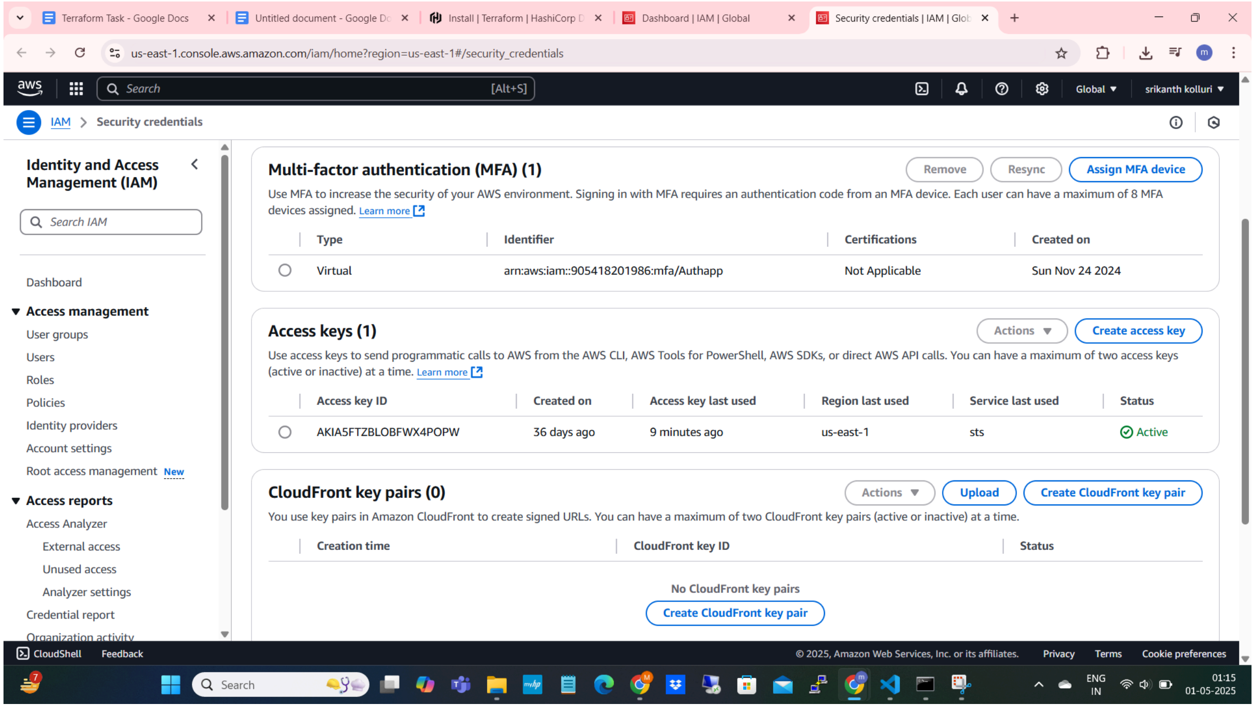


We Need to create Access key and Secret Keys

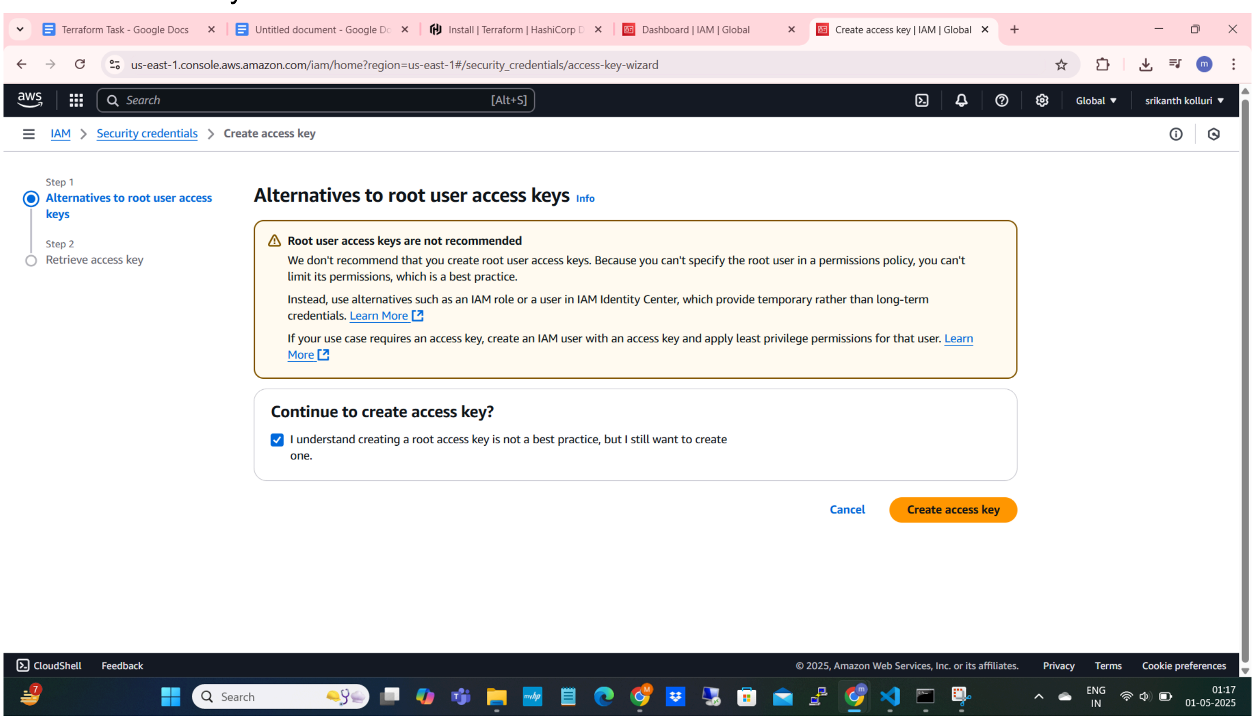
Login to the AWS account and create

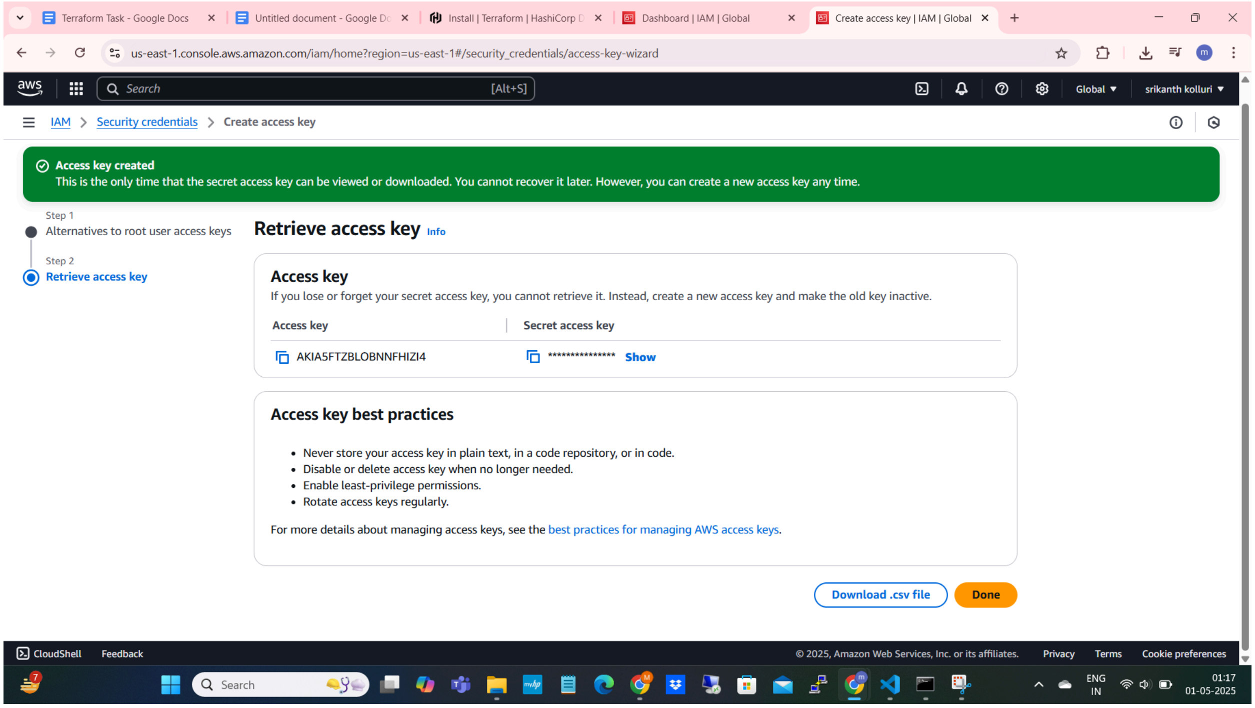
Click on IAM →Manage Access keys



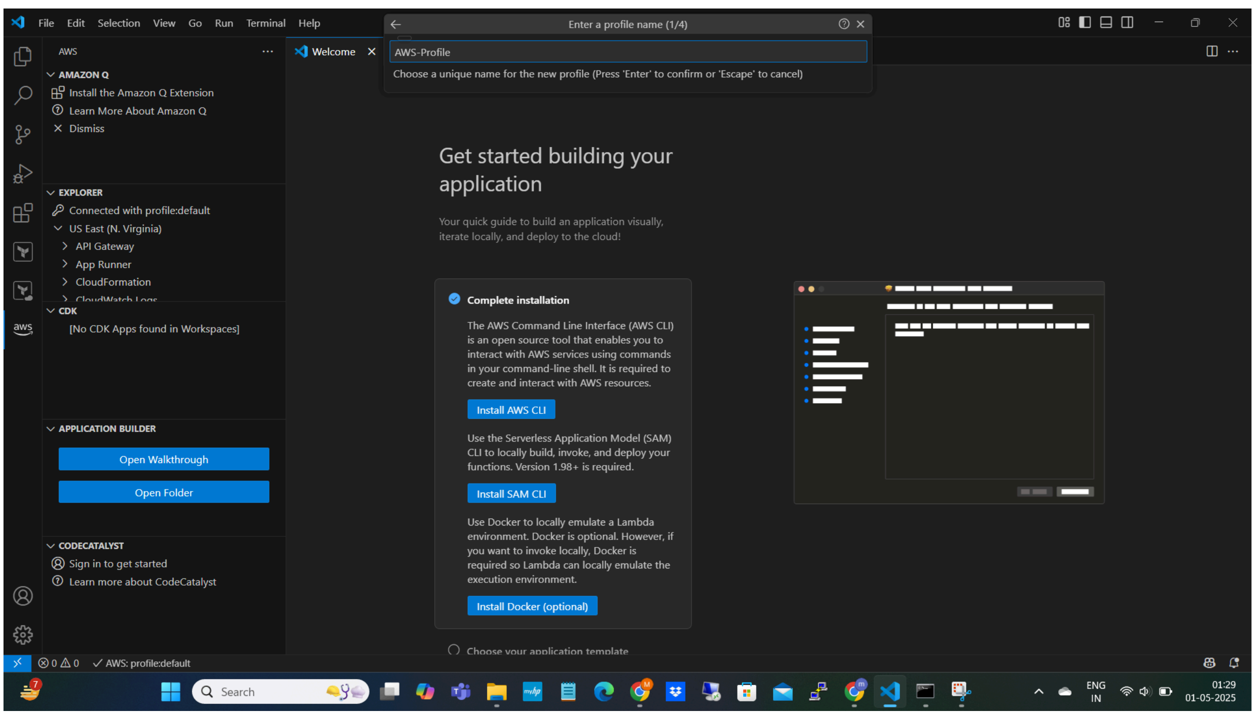


Create Access key

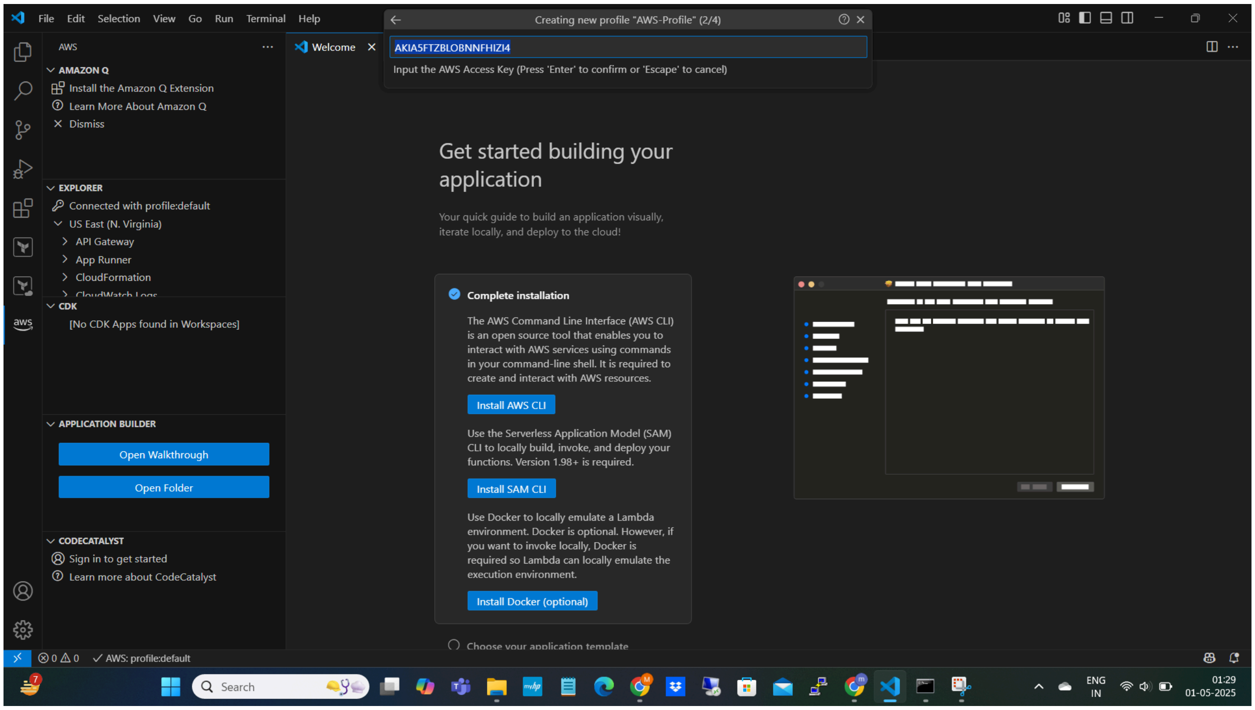




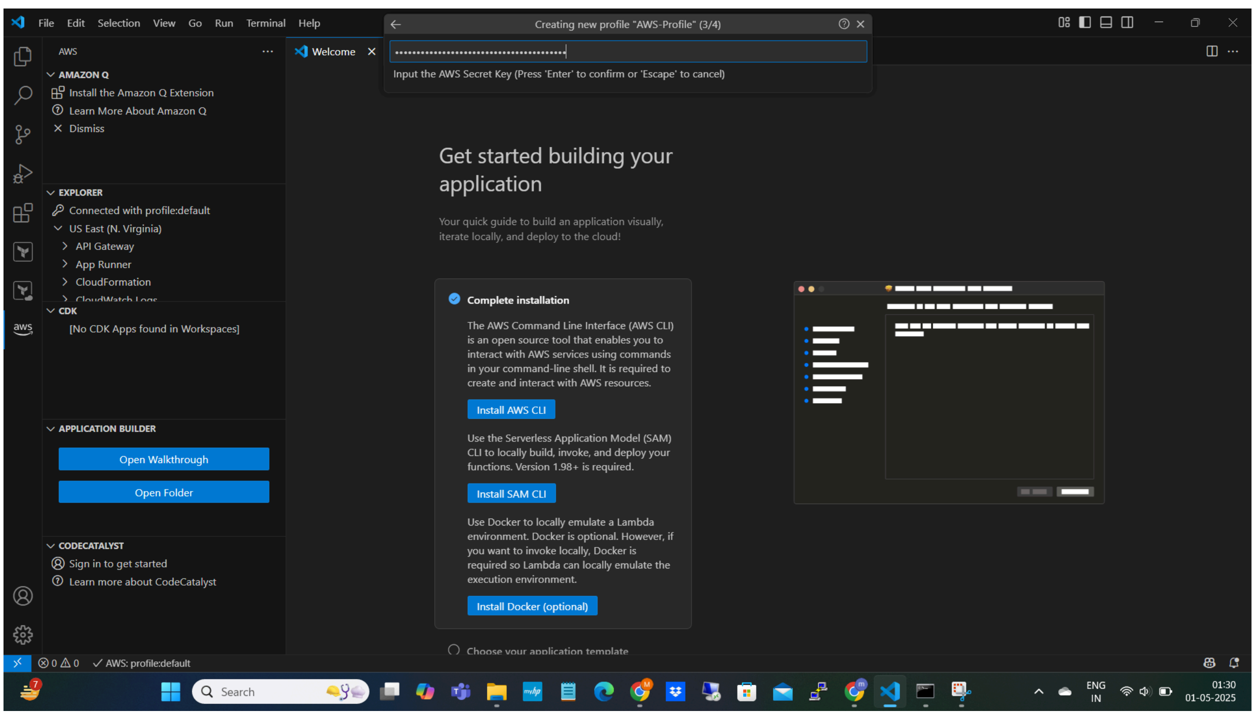
To connect AWS we must configure the access key and Secret key in vscode
Enter Unique name and click Enter

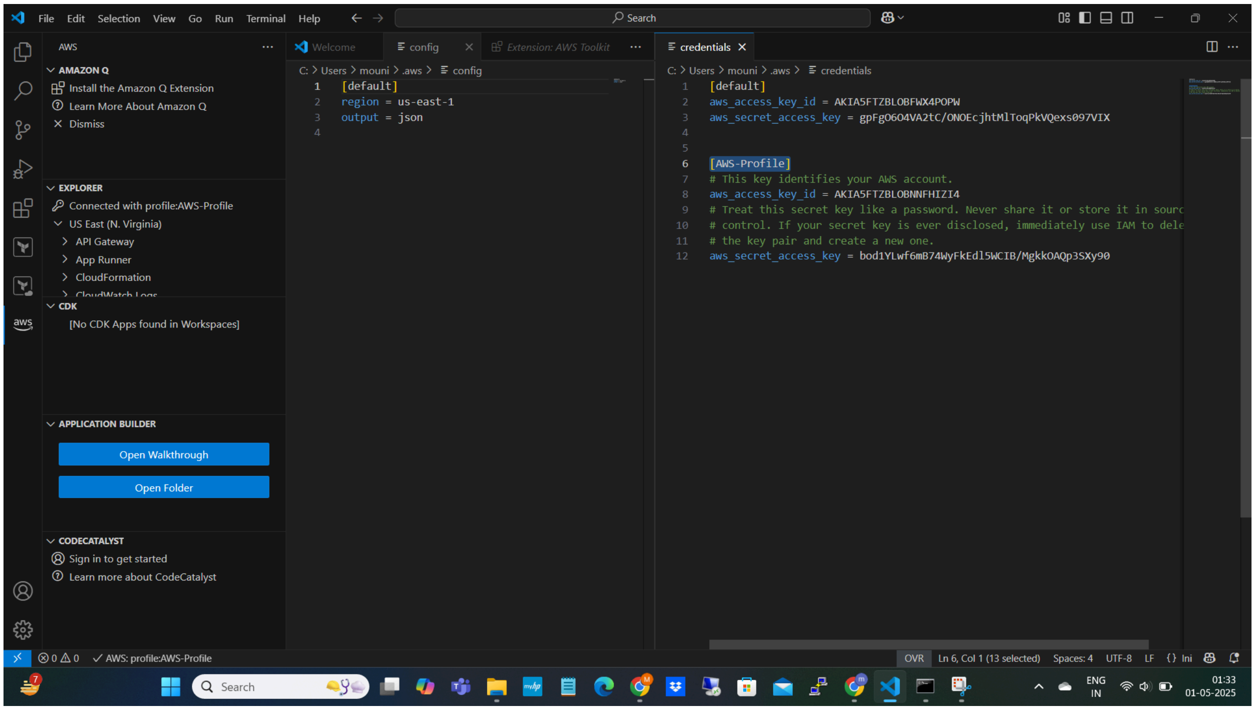


Enter Access key and Click enter



Enter the Secret Access key and click enter

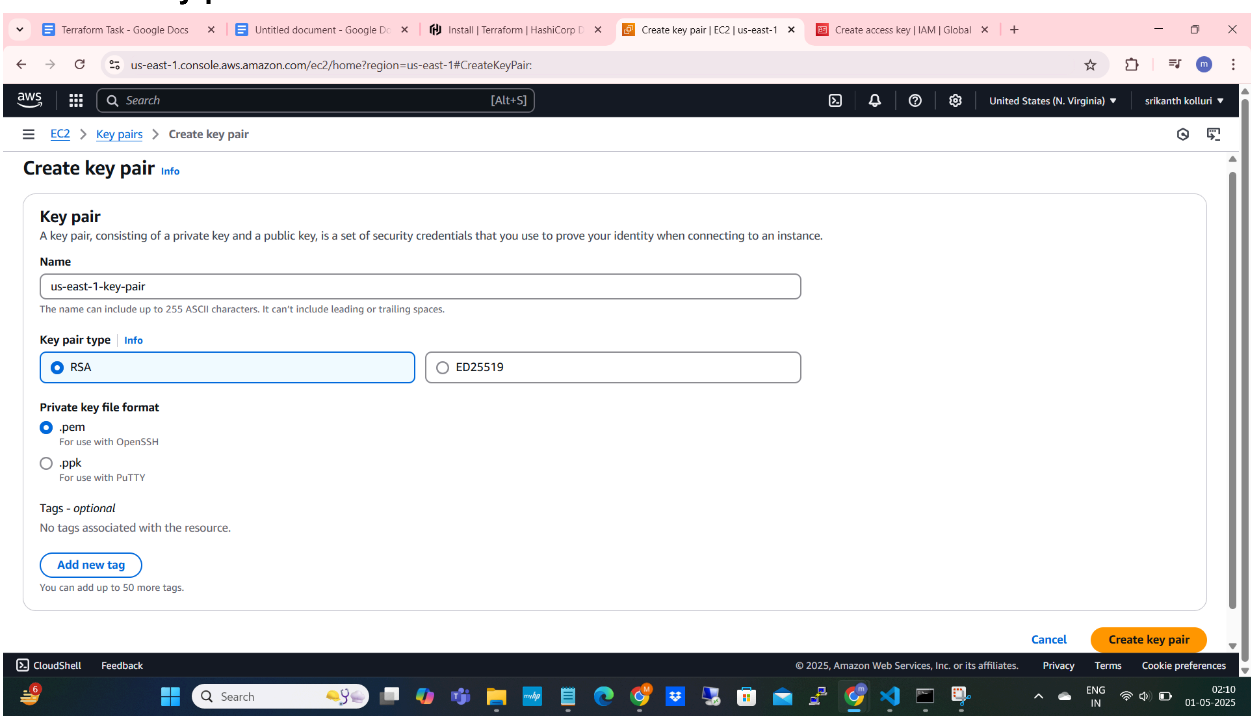


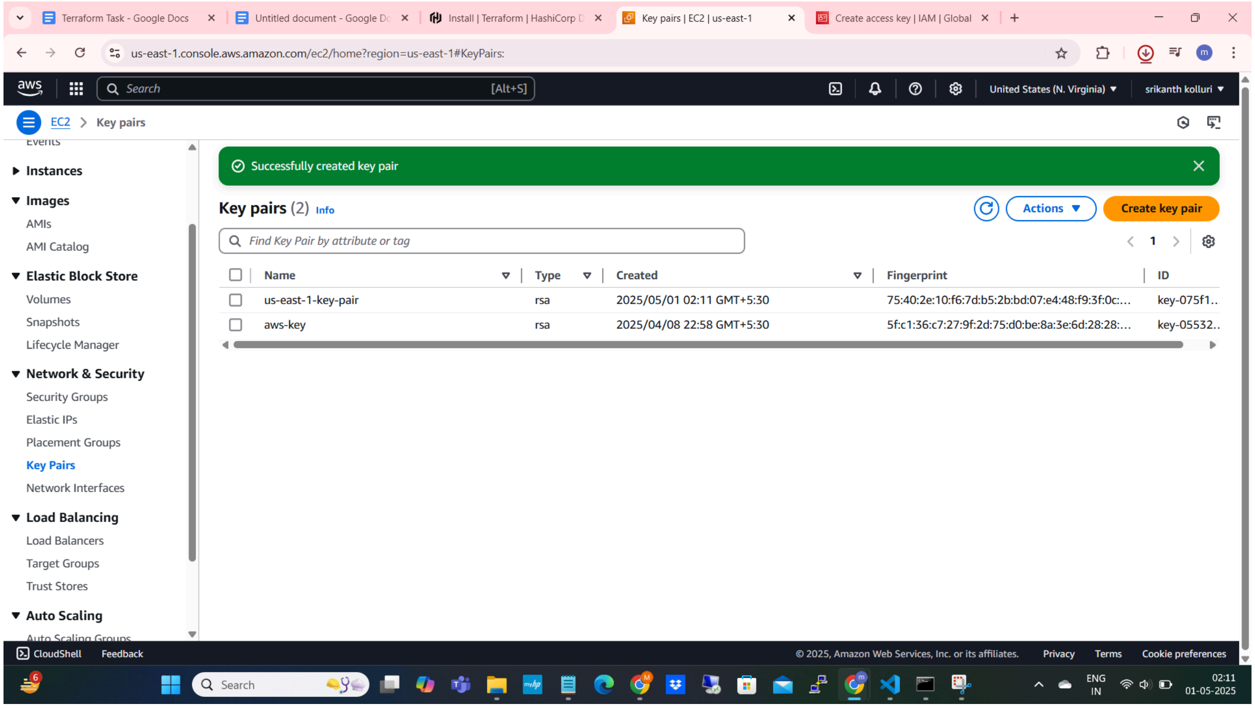


It is Connected Now

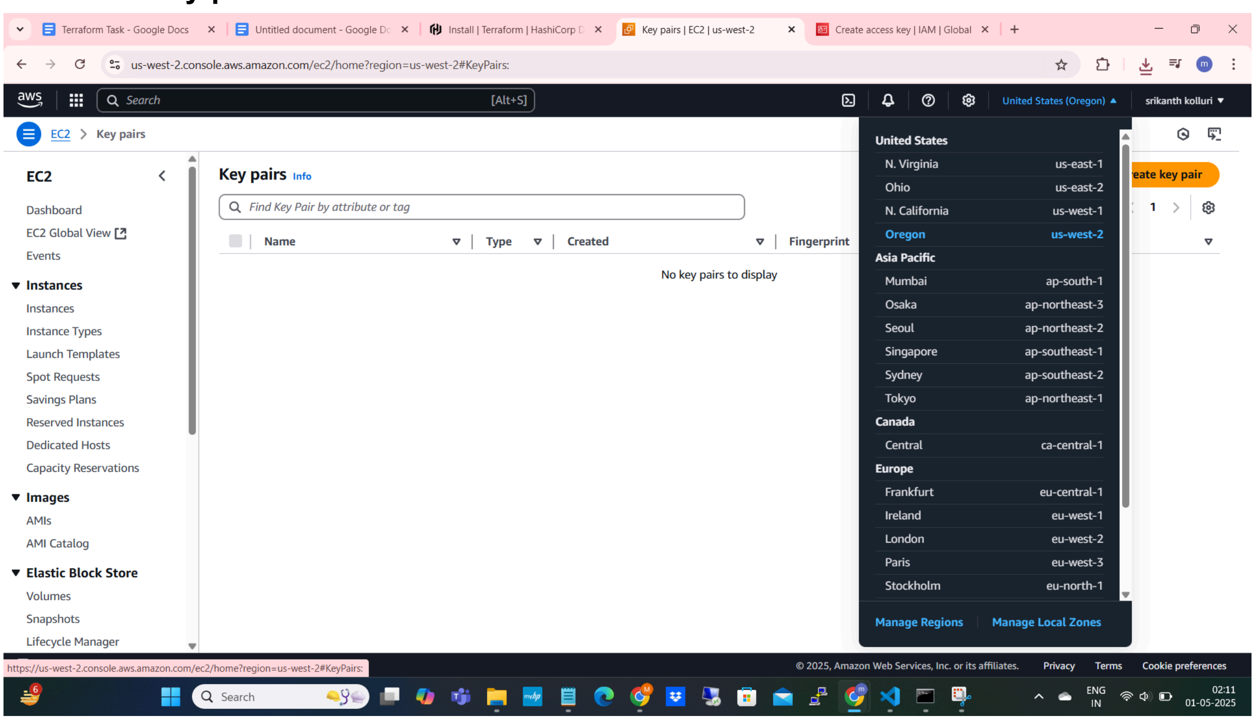
Creating Key pairs in 2 different regions:

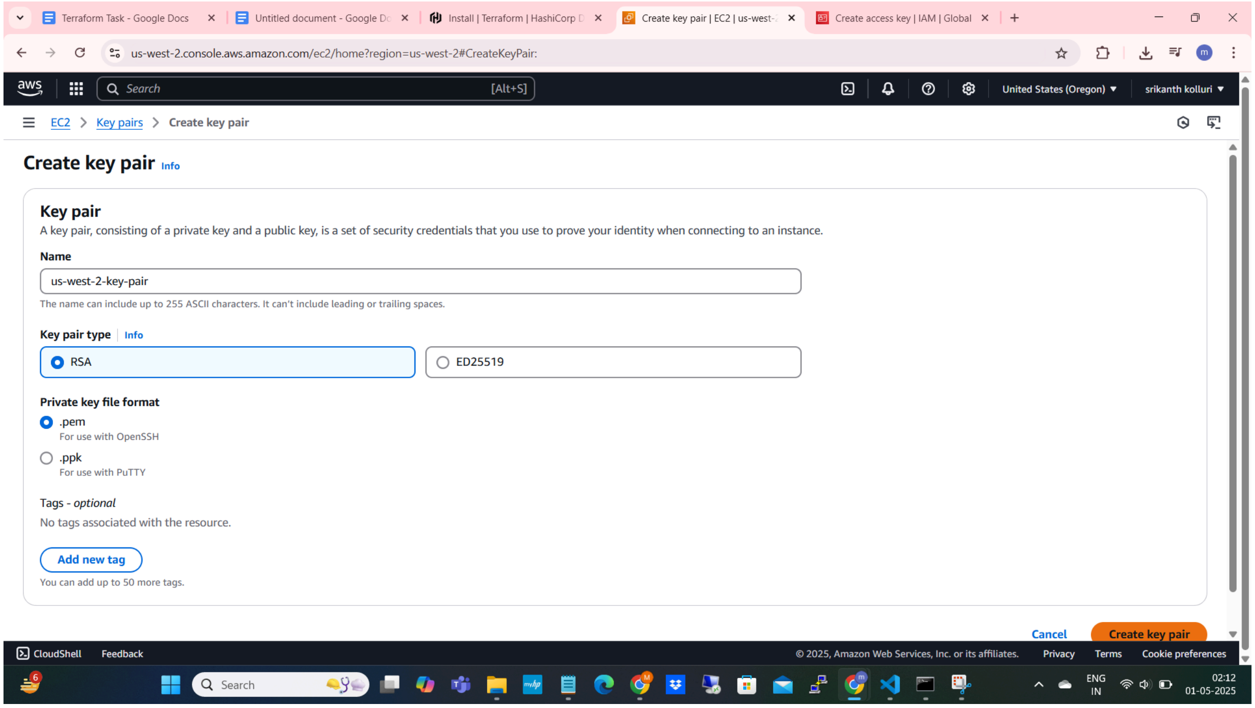
us-east-1-key-pair :

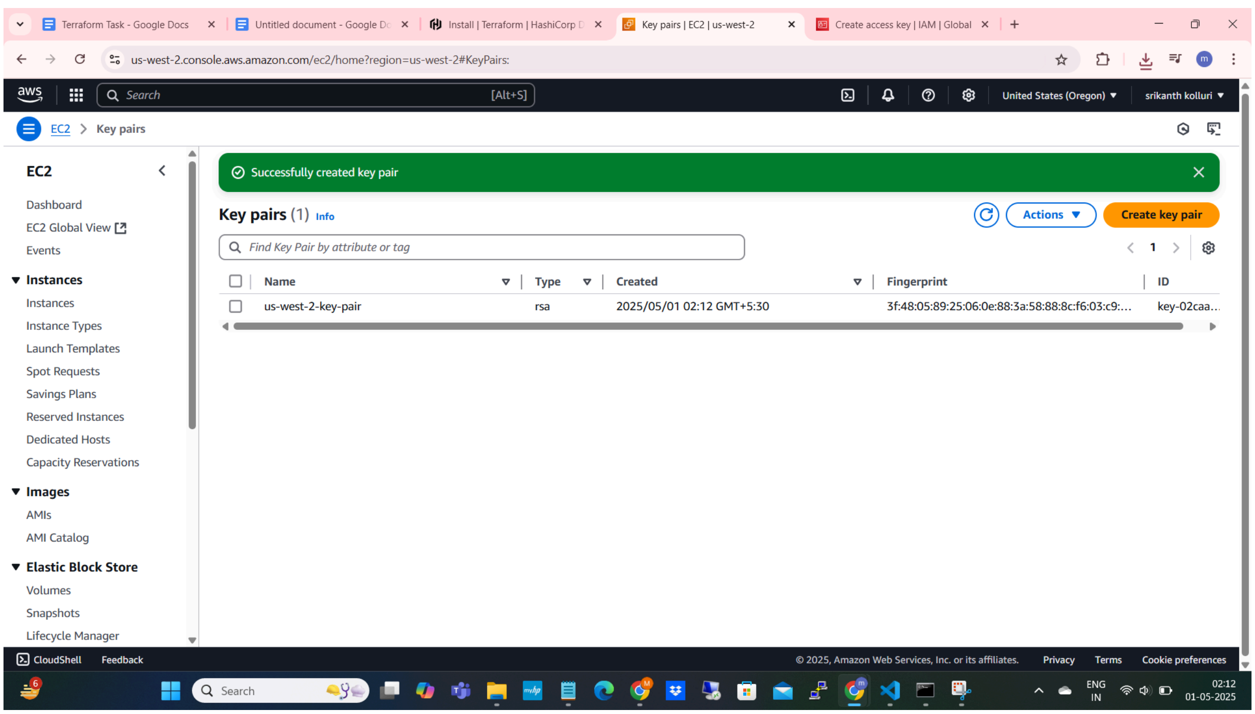




us-west-2-key-pair



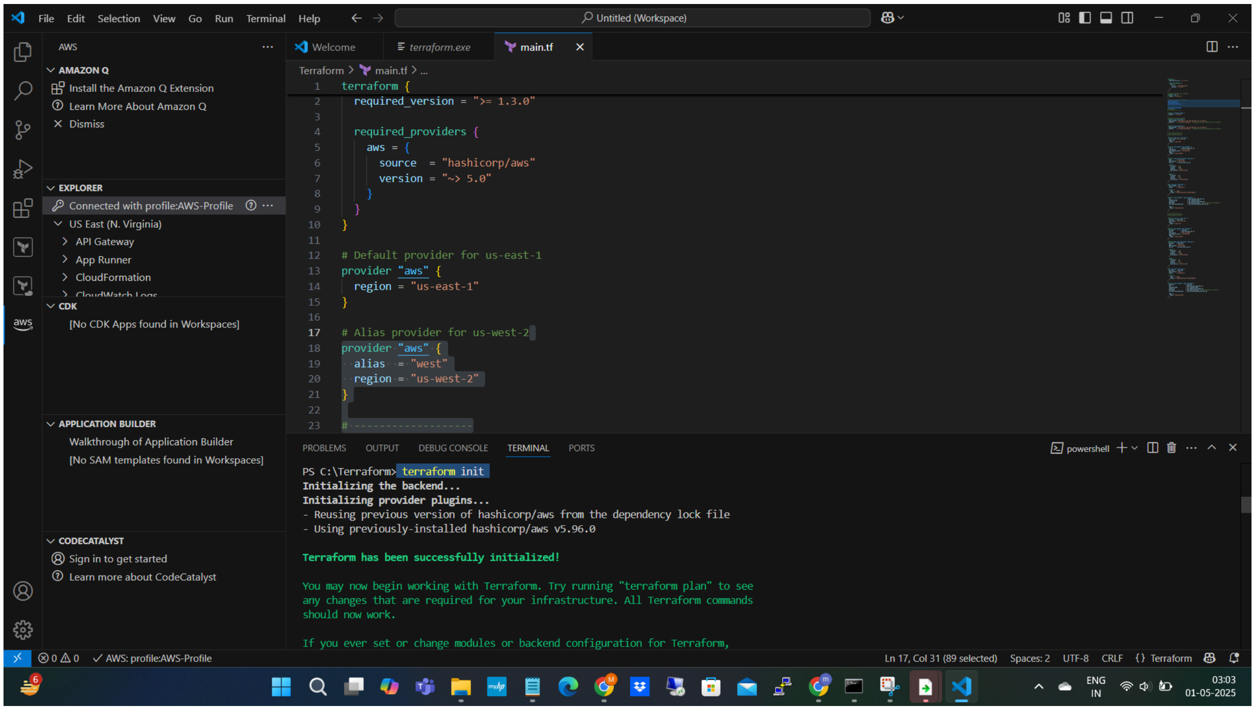

us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#CreateKeyPair:
Create key pair
Key pair
A key pair, consisting of a private key and a public key, is a set of security credentials that you use to prove your identity when connecting to an instance.
Name
us-west-2-key-pair
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.
Key pair type
RSA
ED25519
Private key file format
.pem
For use with OpenSSH
.ppk
For use with PuTTY
Tags - optional
No tags associated with the resource.
Add new tag
You can add up to 50 more tags.
Cancel
Create key pair


us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#KeyPairs:
Successfully created key pair
Key pairs (1)
Actions
Create key pair
Name
Type
Created
Fingerprint
ID
us-west-2-key-pair
rsa
2025/05/01 02:12 GMT+5:30
3f:48:05:89:25:06:0e:88:3a:58:88:8c:f6:03:c9:...
key-02caa...

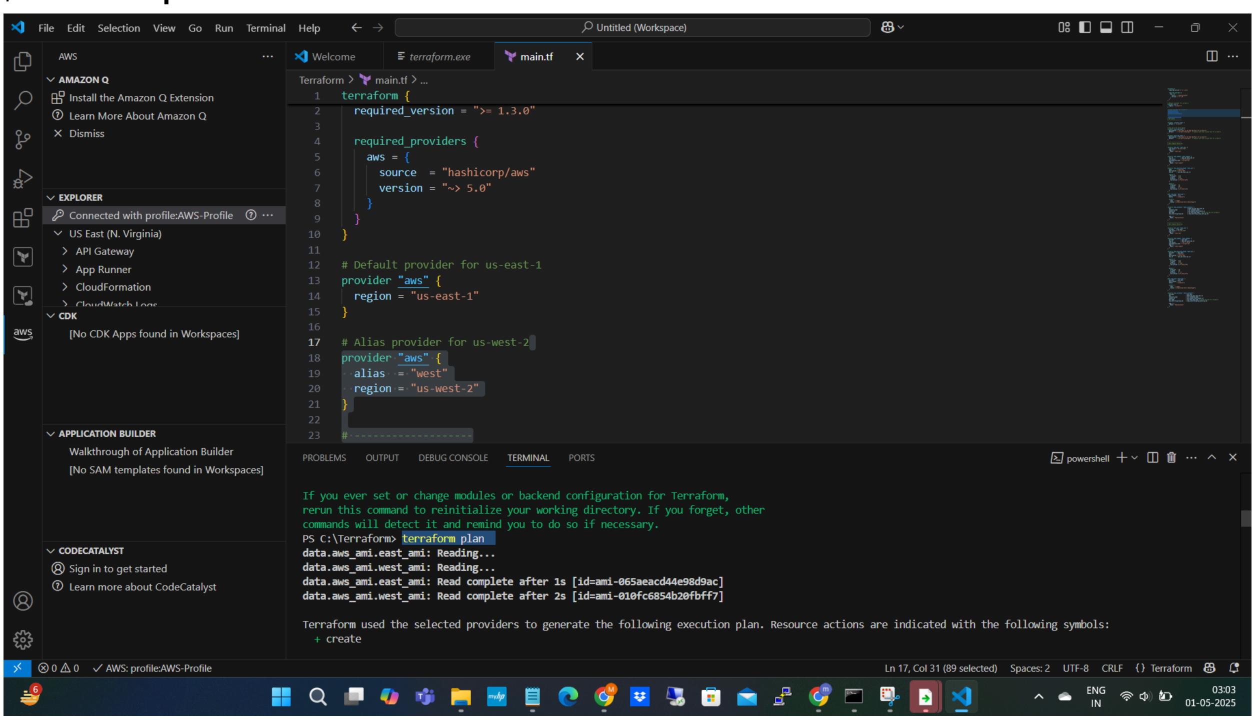
Added main.tf file to create 2 EC2 Instances in 2 Different Regions

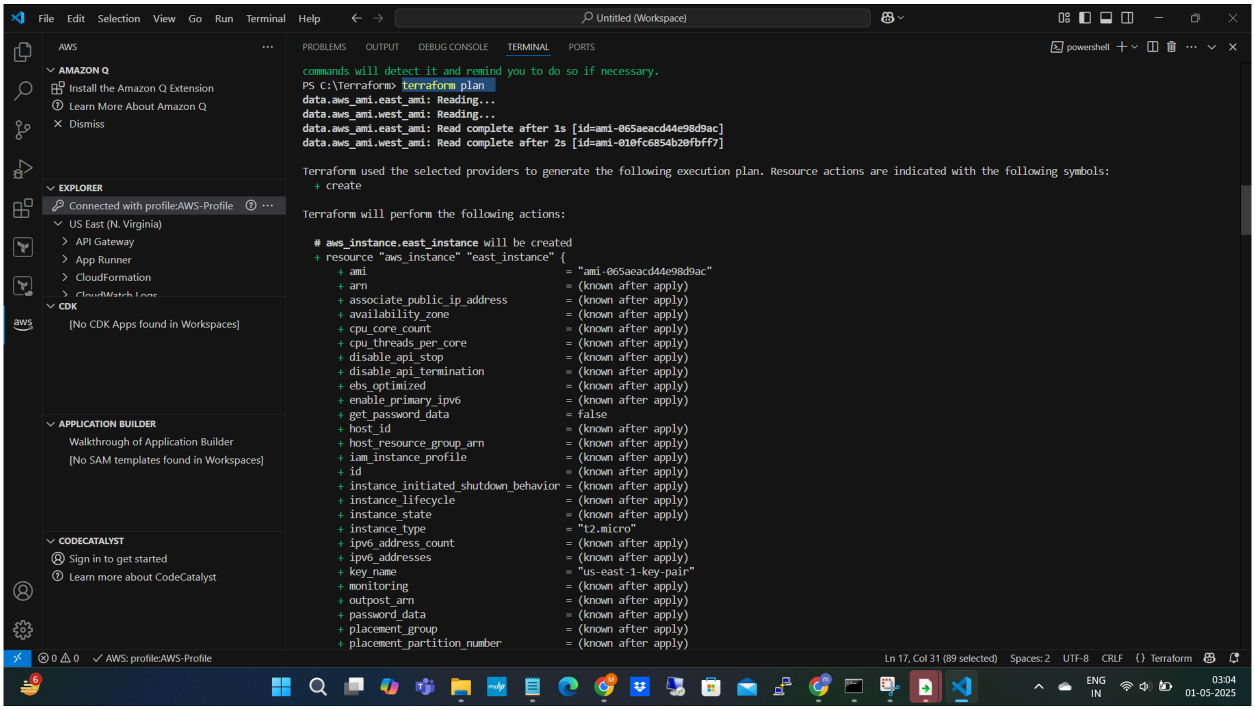
Now initializing the Terraform:

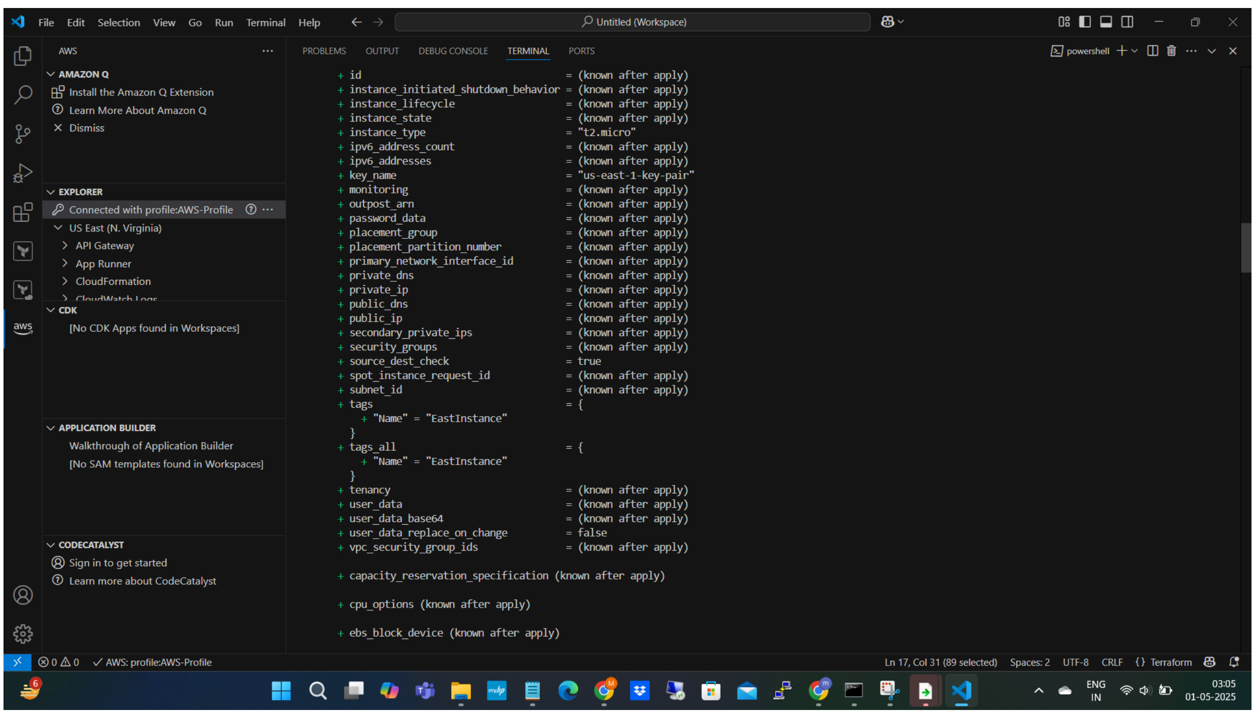
$ terraform init

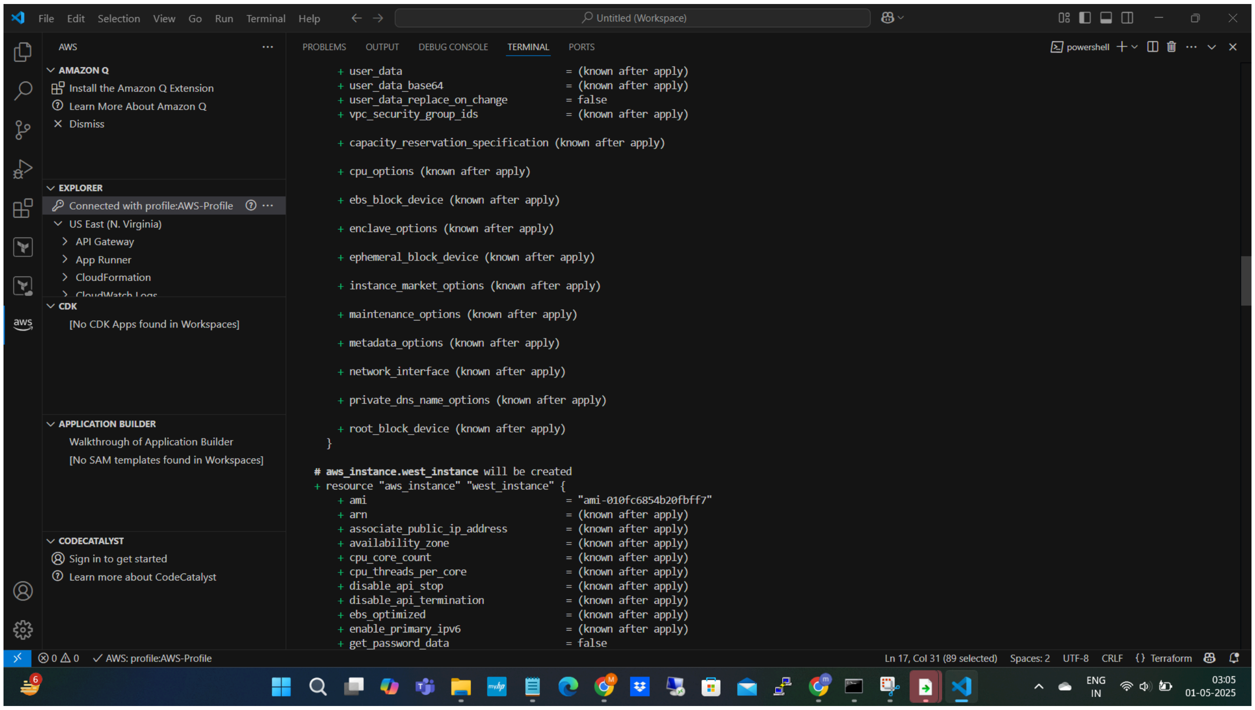


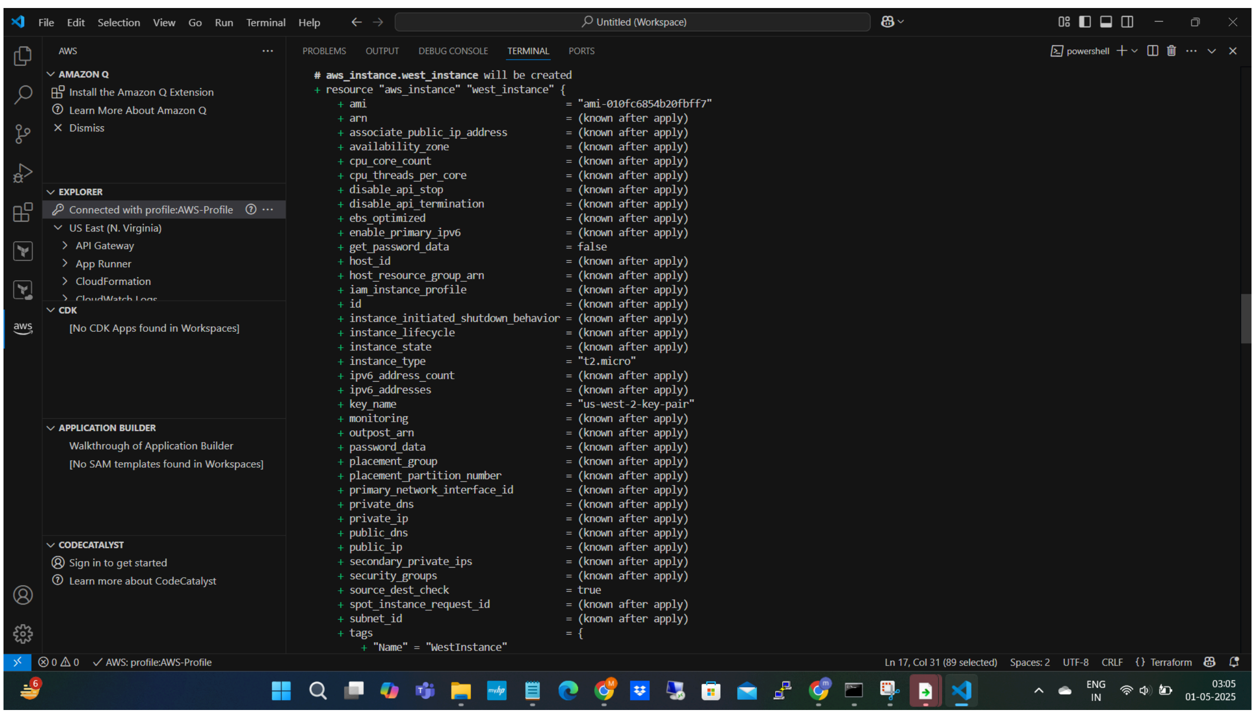
$ terraform plan

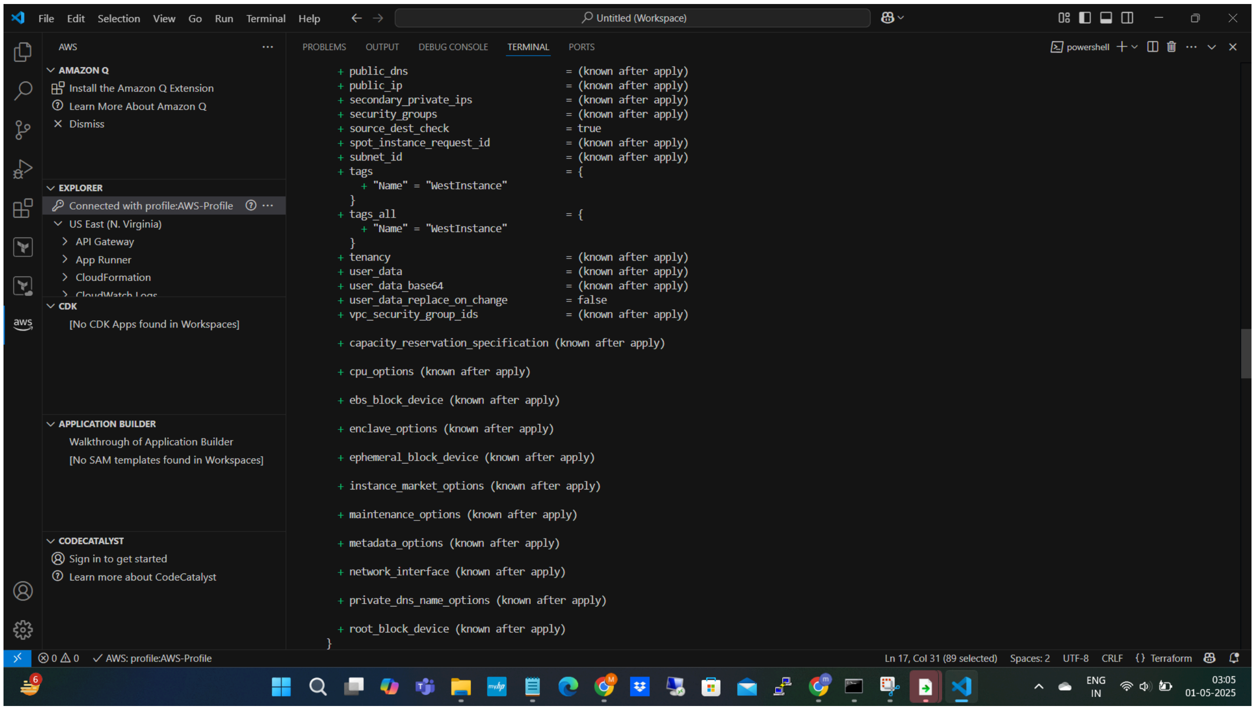


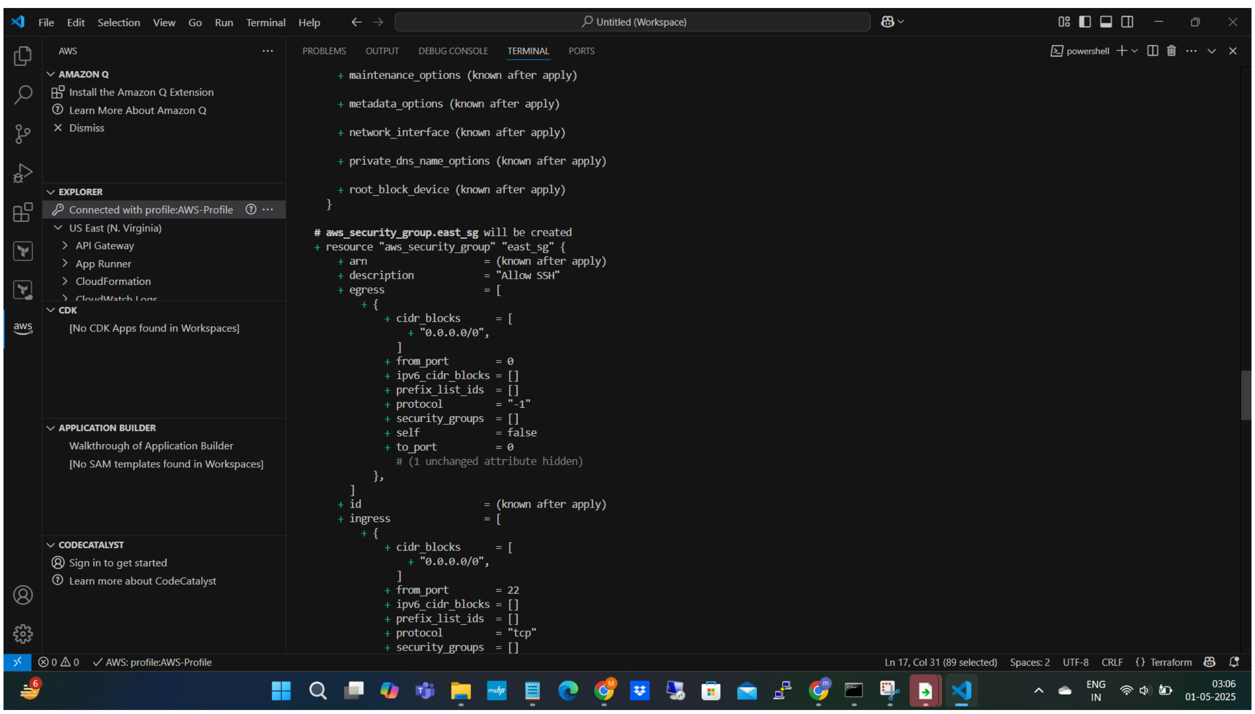

commands will detect it and remind you to do so if necessary.
PS C:\Terraform> terraform plan
data.aws_ami.east_ami: Reading...
data.aws_ami.west_ami: Reading...
data.aws_ami.east_ami: Read complete after 1s [id=ami-065aeacd44e98d9ac]
data.aws_ami.west_ami: Read complete after 2s [id=ami-010fc6854b20fbff7]
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create
Terraform will perform the following actions:
# aws_instance.east_instance will be created
+ resource "aws_instance" "east_instance" {
+ ami = "ami-065aeacd44e98d9ac"
+ arn = (known after apply)
+ associate_public_ip_address = (known after apply)
+ availability_zone = (known after apply)
+ cpu_core_count = (known after apply)
+ cpu_threads_per_core = (known after apply)
+ disable_api_stop = (known after apply)
+ disable_api_termination = (known after apply)
+ ebs_optimized = (known after apply)
+ enable_primary_ipv6 = (known after apply)
+ get_password_data = false
+ host_id = (known after apply)
+ host_resource_group_arn = (known after apply)
+ iam_instance_profile = (known after apply)
+ id = (known after apply)
+ instance_initiated_shutdown_behavior = (known after apply)
+ instance_lifecycle = (known after apply)
+ instance_state = (known after apply)
+ instance_type = "t2.micro"
+ ipv6_address_count = (known after apply)
+ ipv6_addresses = (known after apply)
+ key_name = "us-east-1-key-pair"
+ monitoring = (known after apply)
+ outpost_arn = (known after apply)
+ password_data = (known after apply)
+ placement_group = (known after apply)
+ placement_partition_number = (known after apply)

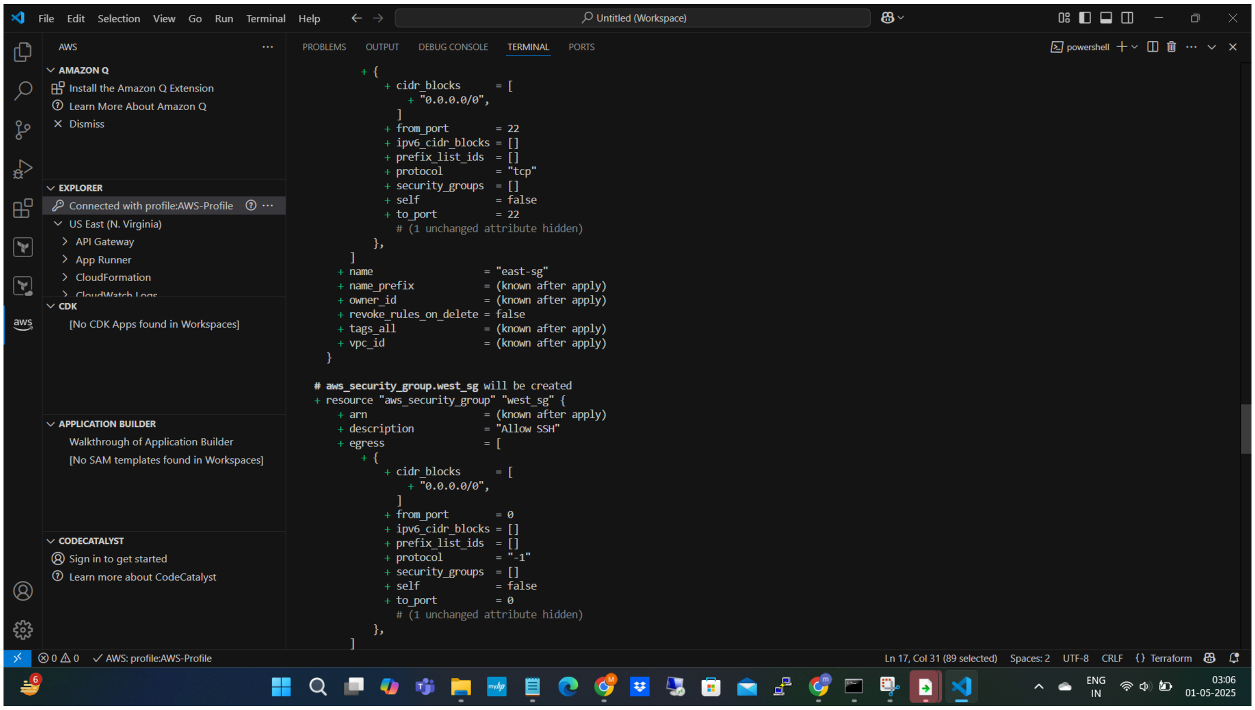

+ id = (known after apply)
+ instance_initiated_shutdown_behavior = (known after apply)
+ instance_lifecycle = (known after apply)
+ instance_state = (known after apply)
+ instance_type = "t2.micro"
+ ipv6_address_count = (known after apply)
+ ipv6_addresses = (known after apply)
+ key_name = "us-east-1-key-pair"
+ monitoring = (known after apply)
+ outpost_arn = (known after apply)
+ password_data = (known after apply)
+ placement_group = (known after apply)
+ placement_partition_number = (known after apply)
+ primary_network_interface_id = (known after apply)
+ private_dns = (known after apply)
+ private_ip = (known after apply)
+ public_dns = (known after apply)
+ public_ip = (known after apply)
+ secondary_private_ips = (known after apply)
+ security_groups = (known after apply)
+ source_dest_check = true
+ spot_instance_request_id = (known after apply)
+ subnet_id = (known after apply)
+ tags = {
+ "Name" = "EastInstance"
}
+ tags_all = {
+ "Name" = "EastInstance"
}
+ tenancy = (known after apply)
+ user_data = (known after apply)
+ user_data_base64 = (known after apply)
+ user_data_replace_on_change = false
+ vpc_security_group_ids = (known after apply)
+ capacity_reservation_specification (known after apply)
+ cpu_options (known after apply)
+ ebs_block_device (known after apply)


File Edit Selection View Go Run Terminal Help
Untitled (Workspace)
AWS
AMAZON Q
Install the Amazon Q Extension
Learn More About Amazon Q
Dismiss
EXPLORER
Connected with profile:AWS-Profile
US East (N. Virginia)
API Gateway
App Runner
CloudFormation
CDK
[No CDK Apps found in Workspaces]
APPLICATION BUILDER
Walkthrough of Application Builder
[No SAM templates found in Workspaces]
CODECATALYST
Sign in to get started
Learn more about CodeCatalyst
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell
+ user_data = (known after apply)
+ user_data_base64 = (known after apply)
+ user_data_replace_on_change = false
+ vpc_security_group_ids = (known after apply)
+ capacity_reservation_specification (known after apply)
+ cpu_options (known after apply)
+ ebs_block_device (known after apply)
+ enclave_options (known after apply)
+ ephemeral_block_device (known after apply)
+ instance_market_options (known after apply)
+ maintenance_options (known after apply)
+ metadata_options (known after apply)
+ network_interface (known after apply)
+ private_dns_name_options (known after apply)
+ root_block_device (known after apply)
}
# aws_instance.west_instance will be created
+ resource "aws_instance" "west_instance" {
+ ami = "ami-010fc6854b20fbff7"
+ arn = (known after apply)
+ associate_public_ip_address = (known after apply)
+ availability_zone = (known after apply)
+ cpu_core_count = (known after apply)
+ cpu_threads_per_core = (known after apply)
+ disable_api_stop = (known after apply)
+ disable_api_termination = (known after apply)
+ ebs_optimized = (known after apply)
+ enable_primary_ipv6 = (known after apply)
+ get_password_data = false
AWS: profile:AWS-Profile
Ln 17, Col 31 (89 selected) Spaces: 2 UTF-8 CRLF Terraform
ENG IN 03:05 01-05-2025

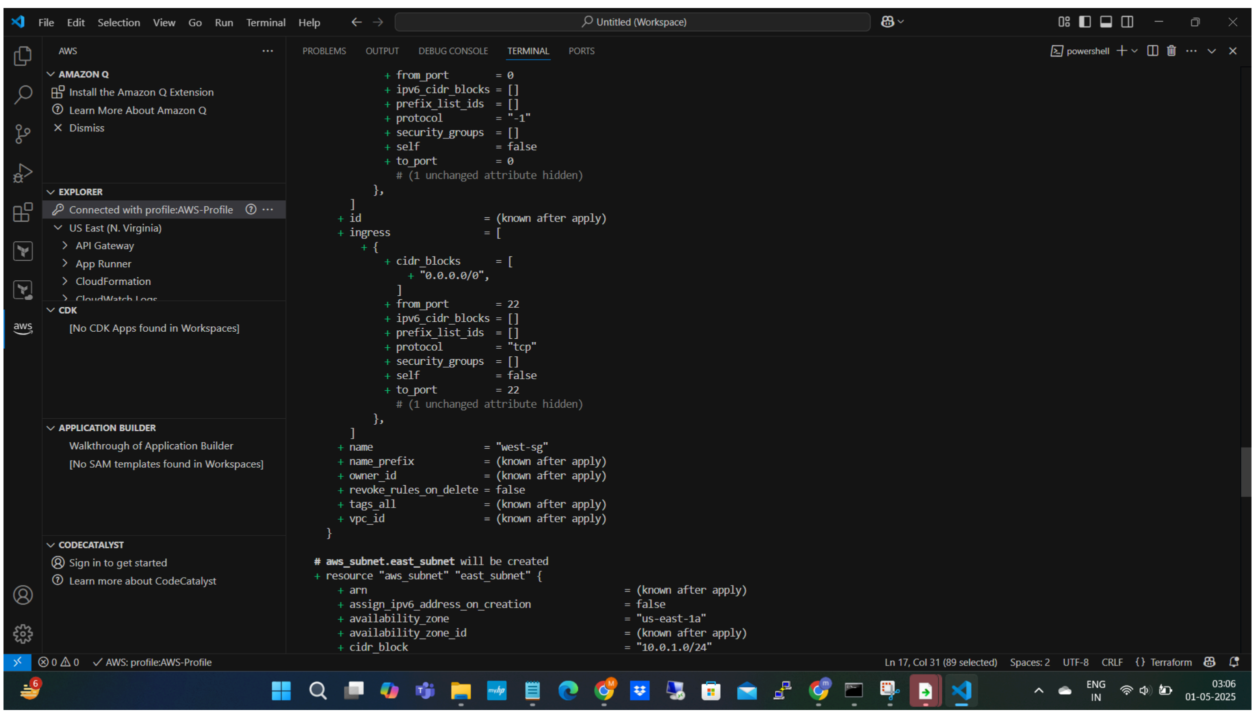

File Edit Selection View Go Run Terminal Help
Untitled (Workspace)
AWS
AMAZON Q
Install the Amazon Q Extension
Learn More About Amazon Q
Dismiss
EXPLORER
Connected with profile:AWS-Profile
US East (N. Virginia)
API Gateway
App Runner
CloudFormation
CDK
[No CDK Apps found in Workspaces]
APPLICATION BUILDER
Walkthrough of Application Builder
[No SAM templates found in Workspaces]
CODECATALYST
Sign in to get started
Learn more about CodeCatalyst
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell
# aws_instance.west_instance will be created
+ resource "aws_instance" "west_instance" {
+ ami = "ami-010fc6854b20fbff7"
+ arn = (known after apply)
+ associate_public_ip_address = (known after apply)
+ availability_zone = (known after apply)
+ cpu_core_count = (known after apply)
+ cpu_threads_per_core = (known after apply)
+ disable_api_stop = (known after apply)
+ disable_api_termination = (known after apply)
+ ebs_optimized = (known after apply)
+ enable_primary_ipv6 = (known after apply)
+ get_password_data = false
+ host_id = (known after apply)
+ host_resource_group_arn = (known after apply)
+ iam_instance_profile = (known after apply)
+ id = (known after apply)
+ instance_initiated_shutdown_behavior = (known after apply)
+ instance_lifecycle = (known after apply)
+ instance_state = (known after apply)
+ instance_type = "t2.micro"
+ ipv6_address_count = (known after apply)
+ ipv6_addresses = (known after apply)
+ key_name = "us-west-2-key-pair"
+ monitoring = (known after apply)
+ outpost_arn = (known after apply)
+ password_data = (known after apply)
+ placement_group = (known after apply)
+ placement_partition_number = (known after apply)
+ primary_network_interface_id = (known after apply)
+ private_dns = (known after apply)
+ private_ip = (known after apply)
+ public_dns = (known after apply)
+ public_ip = (known after apply)
+ secondary_private_ips = (known after apply)
+ security_groups = (known after apply)
+ source_dest_check = true
+ spot_instance_request_id = (known after apply)
+ subnet_id = (known after apply)
+ tags = {
+ "Name" = "WestInstance"
AWS: profile:AWS-Profile
Ln 17, Col 31 (89 selected) Spaces: 2 UTF-8 CRLF Terraform
ENG IN 03:05 01-05-2025


+ public_dns = (known after apply)
+ public_ip = (known after apply)
+ secondary_private_ips = (known after apply)
+ security_groups = (known after apply)
+ source_dest_check = true
+ spot_instance_request_id = (known after apply)
+ subnet_id = (known after apply)
+ tags = {
+ "Name" = "WestInstance"
}
+ tags_all = {
+ "Name" = "WestInstance"
}
+ tenancy = (known after apply)
+ user_data = (known after apply)
+ user_data_base64 = (known after apply)
+ user_data_replace_on_change = false
+ vpc_security_group_ids = (known after apply)
+ capacity_reservation_specification (known after apply)
+ cpu_options (known after apply)
+ ebs_block_device (known after apply)
+ enclave_options (known after apply)
+ ephemeral_block_device (known after apply)
+ instance_market_options (known after apply)
+ maintenance_options (known after apply)
+ metadata_options (known after apply)
+ network_interface (known after apply)
+ private_dns_name_options (known after apply)
+ root_block_device (known after apply)
}

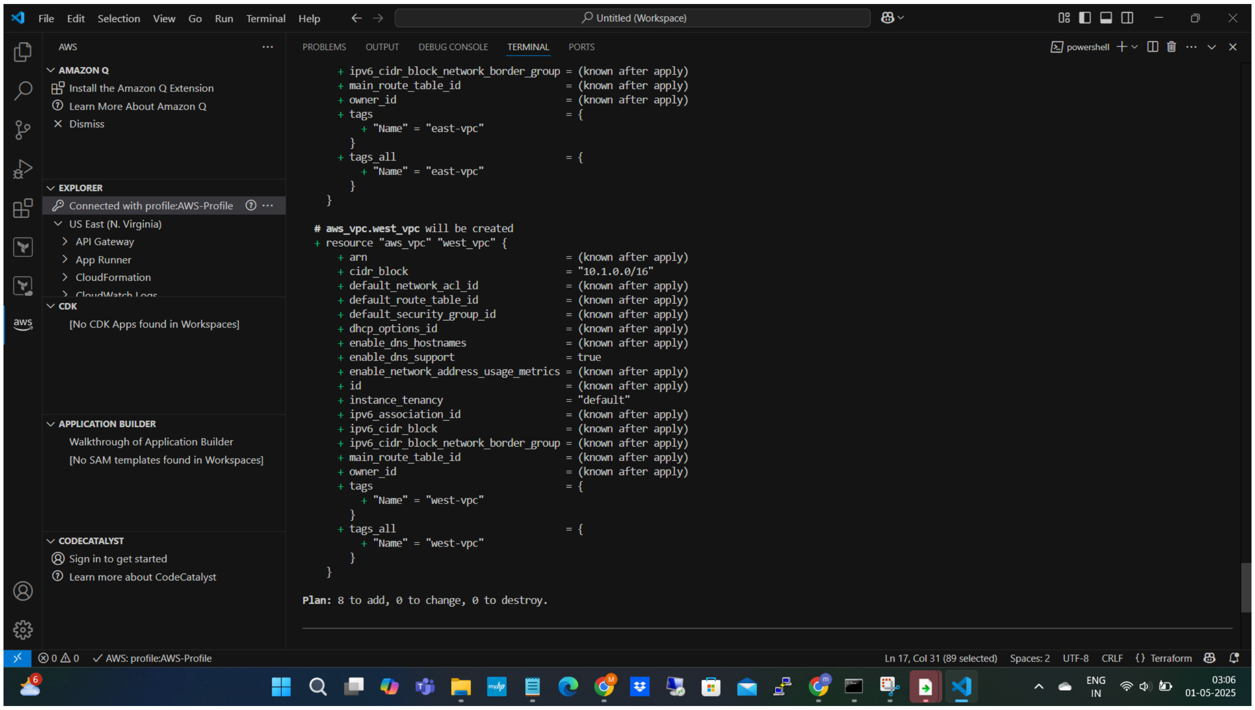

+ maintenance_options (known after apply)
+ metadata_options (known after apply)
+ network_interface (known after apply)
+ private_dns_name_options (known after apply)
+ root_block_device (known after apply)
}
# aws_security_group.east_sg will be created
+ resource "aws_security_group" "east_sg" {
+ arn = (known after apply)
+ description = "Allow SSH"
+ egress = [
+ {
+ cidr_blocks = [
+ "0.0.0.0/0",
]
+ from_port = 0
+ ipv6_cidr_blocks = []
+ prefix_list_ids = []
+ protocol = "-1"
+ security_groups = []
+ self = false
+ to_port = 0
# (1 unchanged attribute hidden)
},
]
+ id = (known after apply)
+ ingress = [
+ {
+ cidr_blocks = [
+ "0.0.0.0/0",
]
+ from_port = 22
+ ipv6_cidr_blocks = []
+ prefix_list_ids = []
+ protocol = "tcp"
+ security_groups = []



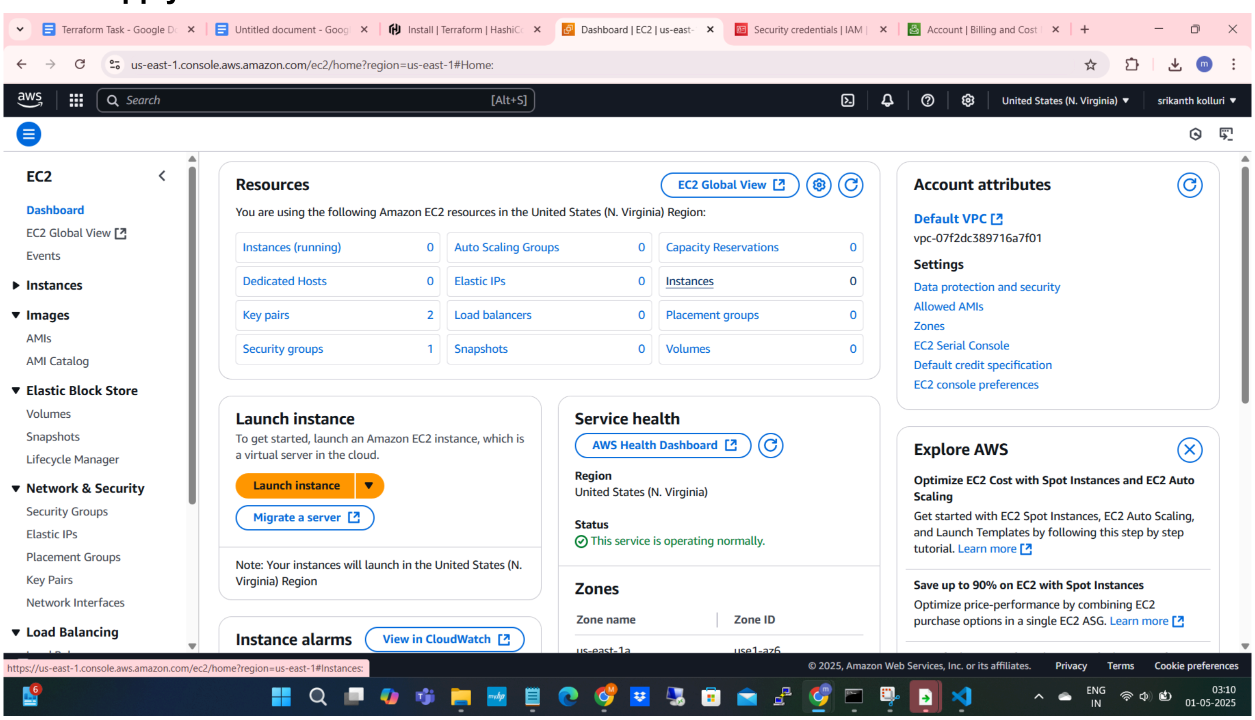
+ {
+ cidr_blocks = [
+ "0.0.0.0/0",
]
+ from_port = 22
+ ipv6_cidr_blocks = []
+ prefix_list_ids = []
+ protocol = "tcp"
+ security_groups = []
+ self = false
+ to_port = 22
# (1 unchanged attribute hidden)
},
]
+ name = "east-sg"
+ name_prefix = (known after apply)
+ owner_id = (known after apply)
+ revoke_rules_on_delete = false
+ tags_all = (known after apply)
+ vpc_id = (known after apply)
}
# aws_security_group.west_sg will be created
+ resource "aws_security_group" "west_sg" {
+ arn = (known after apply)
+ description = "Allow SSH"
+ egress = [
+ {
+ cidr_blocks = [
+ "0.0.0.0/0",
]
+ from_port = 0
+ ipv6_cidr_blocks = []
+ prefix_list_ids = []
+ protocol = "-1"
+ security_groups = []
+ self = false
+ to_port = 0
# (1 unchanged attribute hidden)
},
]



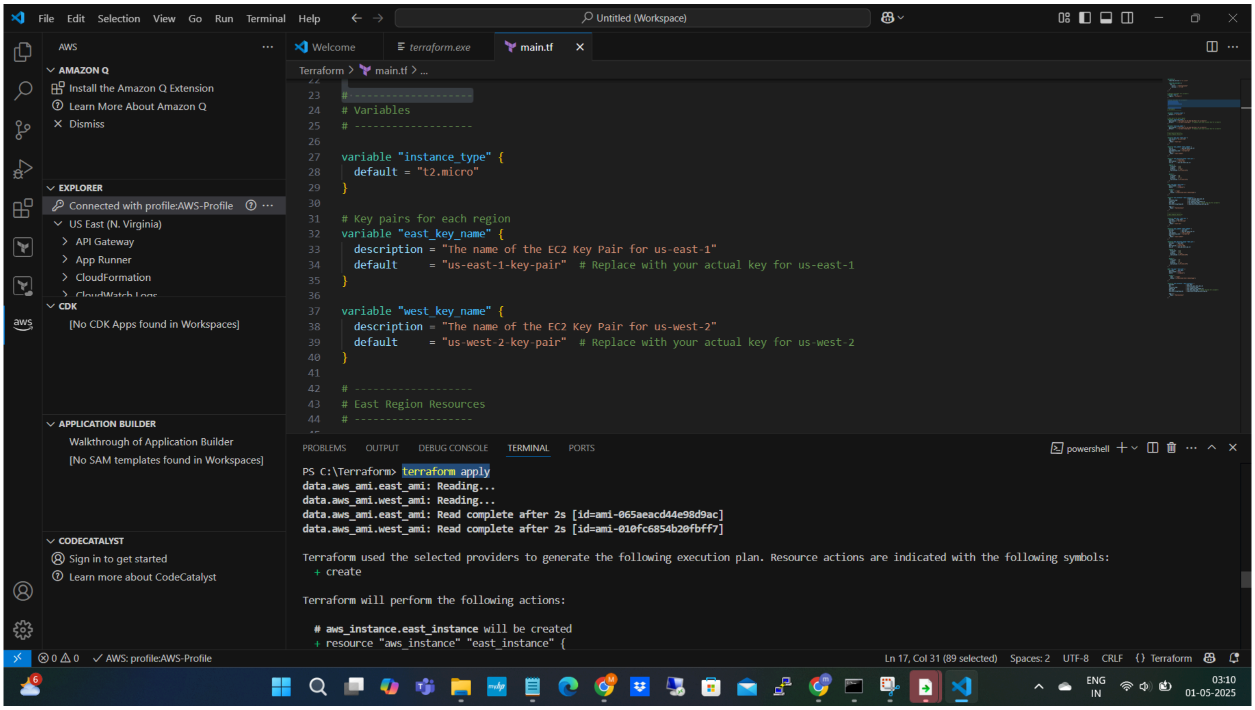
+ from_port = 0
+ ipv6_cidr_blocks = []
+ prefix_list_ids = []
+ protocol = "-1"
+ security_groups = []
+ self = false
+ to_port = 0
# (1 unchanged attribute hidden)
},
]
+ id = (known after apply)
+ ingress = [
+ {
+ cidr_blocks = [
+ "0.0.0.0/0",
]
+ from_port = 22
+ ipv6_cidr_blocks = []
+ prefix_list_ids = []
+ protocol = "tcp"
+ security_groups = []
+ self = false
+ to_port = 22
# (1 unchanged attribute hidden)
},
]
+ name = "west-sg"
+ name_prefix = (known after apply)
+ owner_id = (known after apply)
+ revoke_rules_on_delete = false
+ tags_all = (known after apply)
+ vpc_id = (known after apply)
}
# aws_subnet.east_subnet will be created
+ resource "aws_subnet" "east_subnet" {
+ arn = (known after apply)
+ assign_ipv6_address_on_creation = false
+ availability_zone = "us-east-1a"
+ availability_zone_id = (known after apply)
+ cidr_block = "10.0.1.0/24"

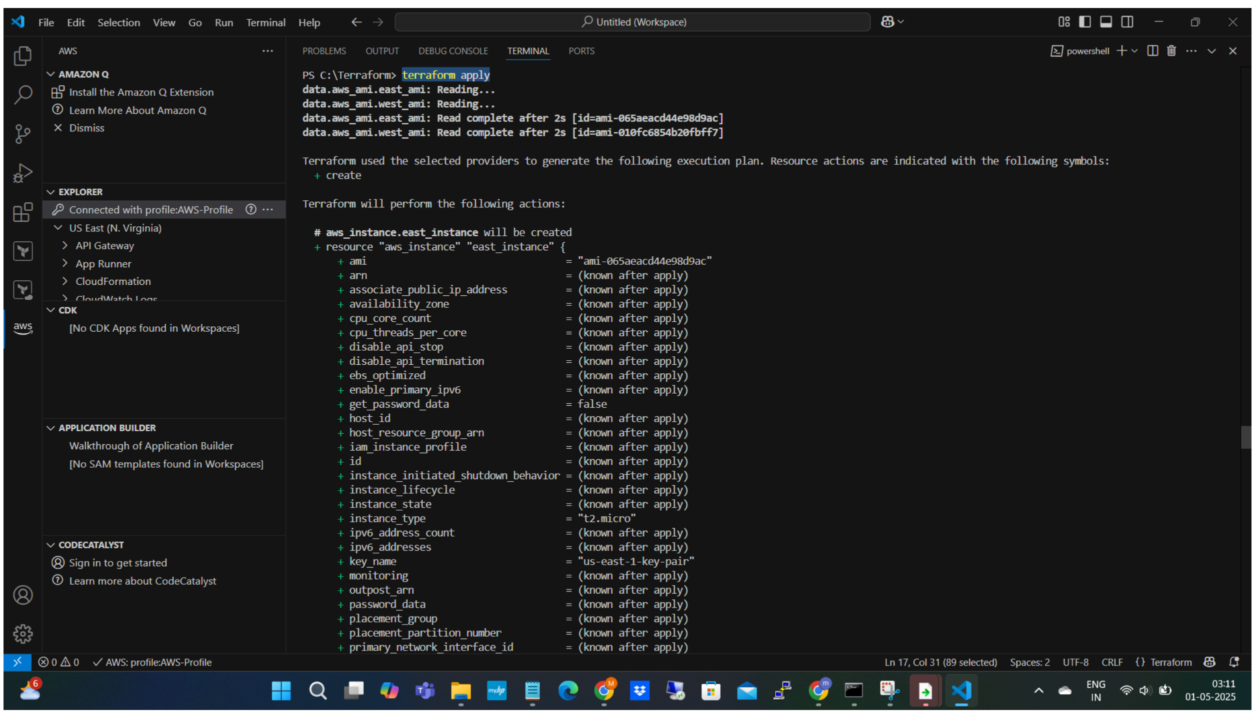


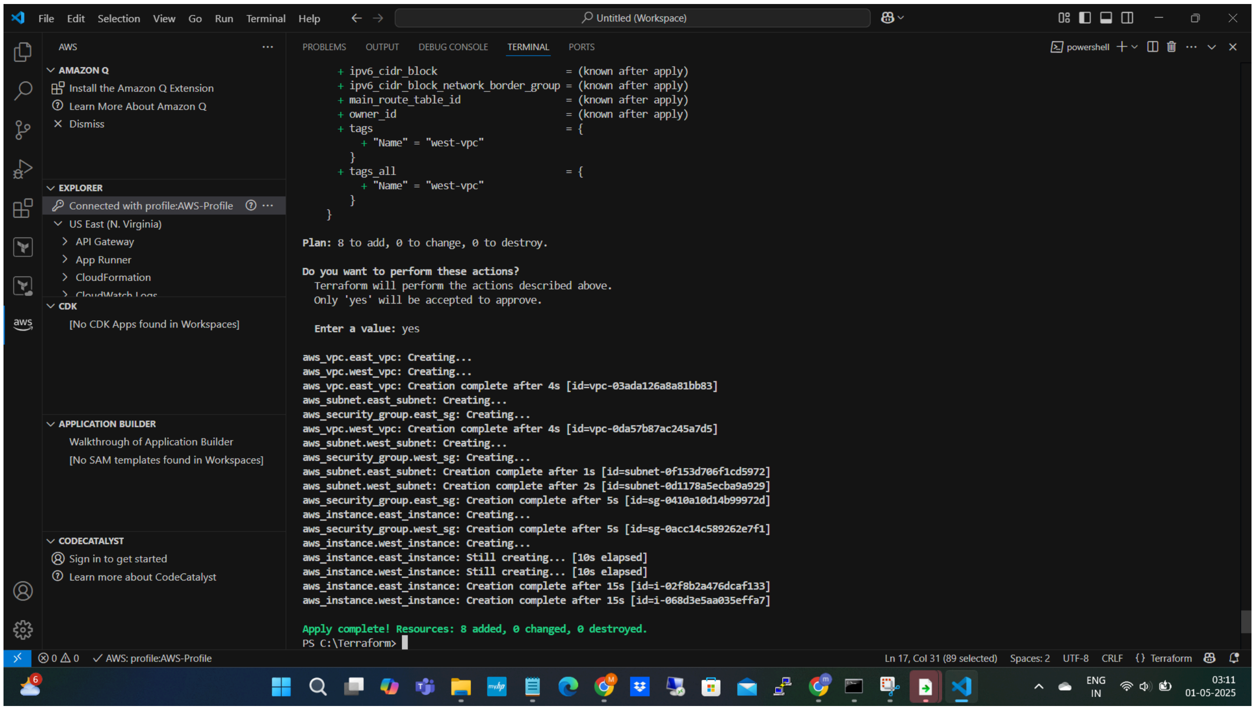
Before Apply- No Ec2 Instances



$ terraform apply



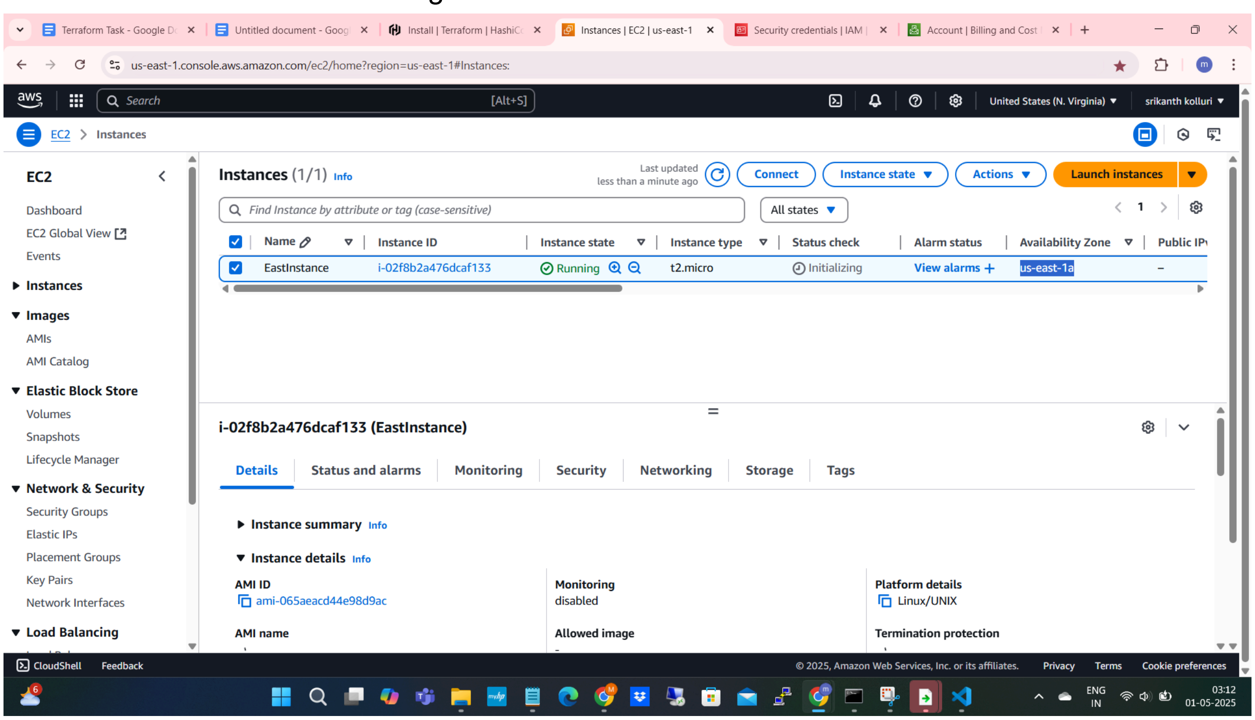


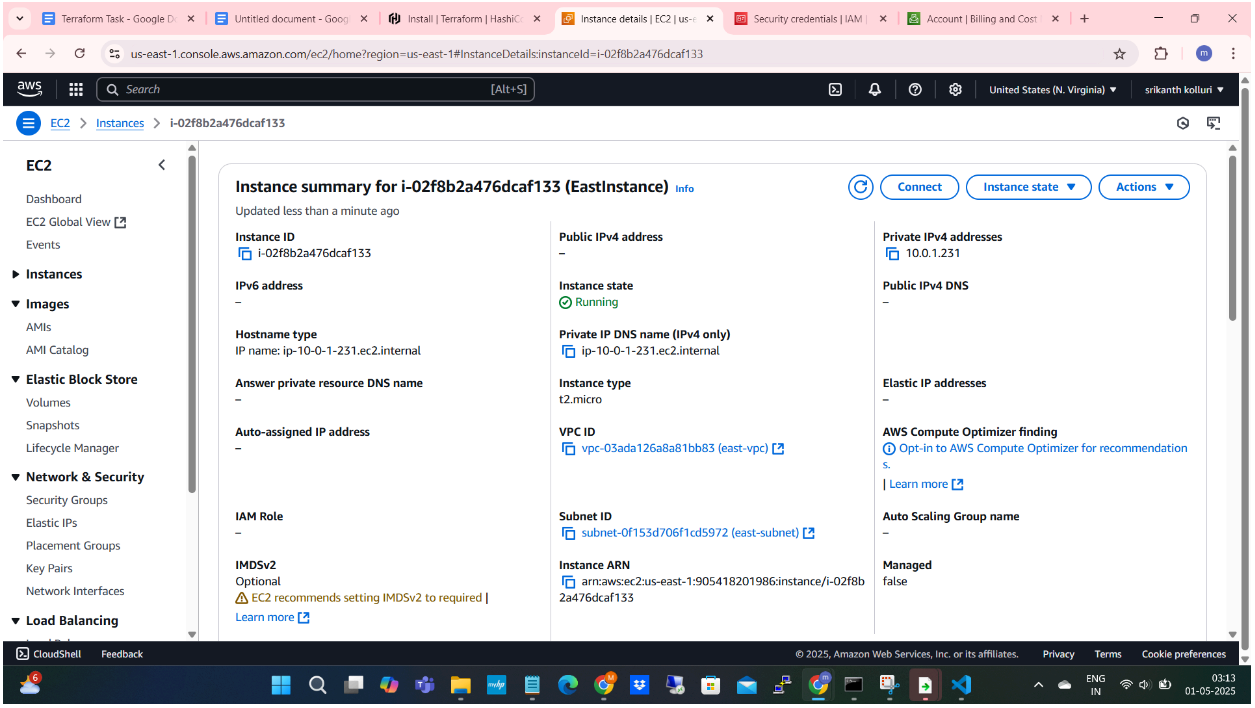


Successfully applied the terraform

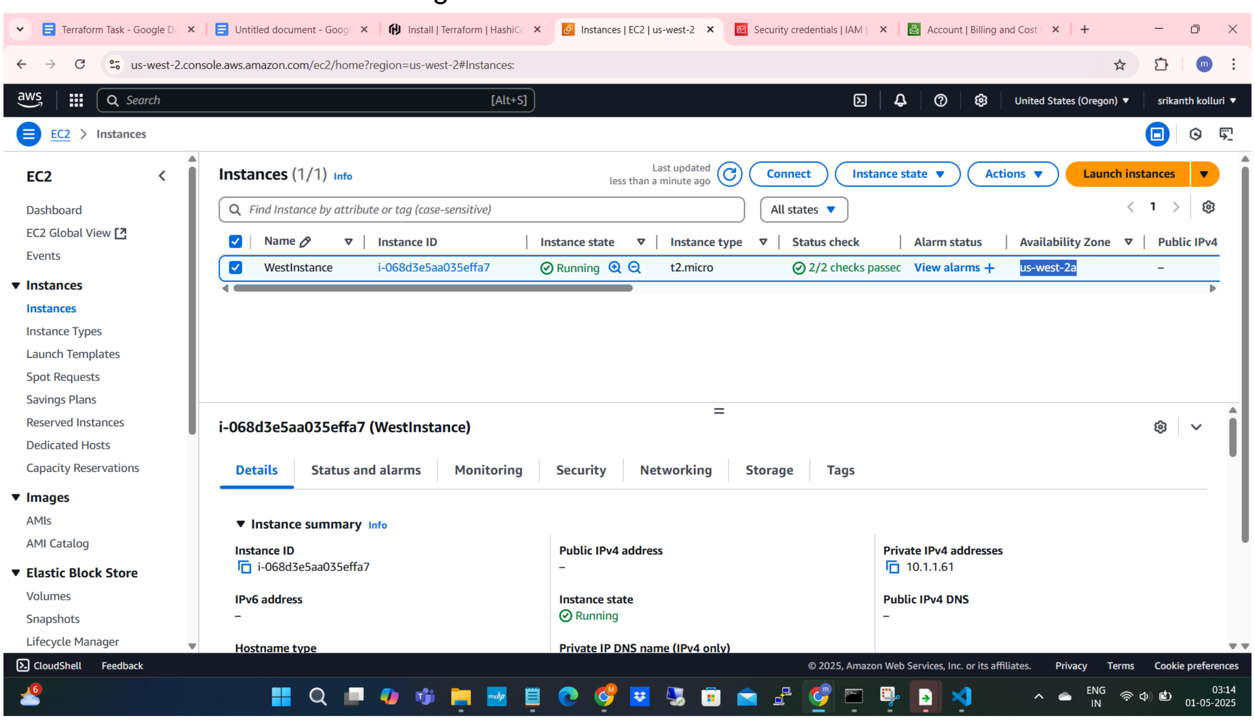
Check AWS whether the instances are created or not

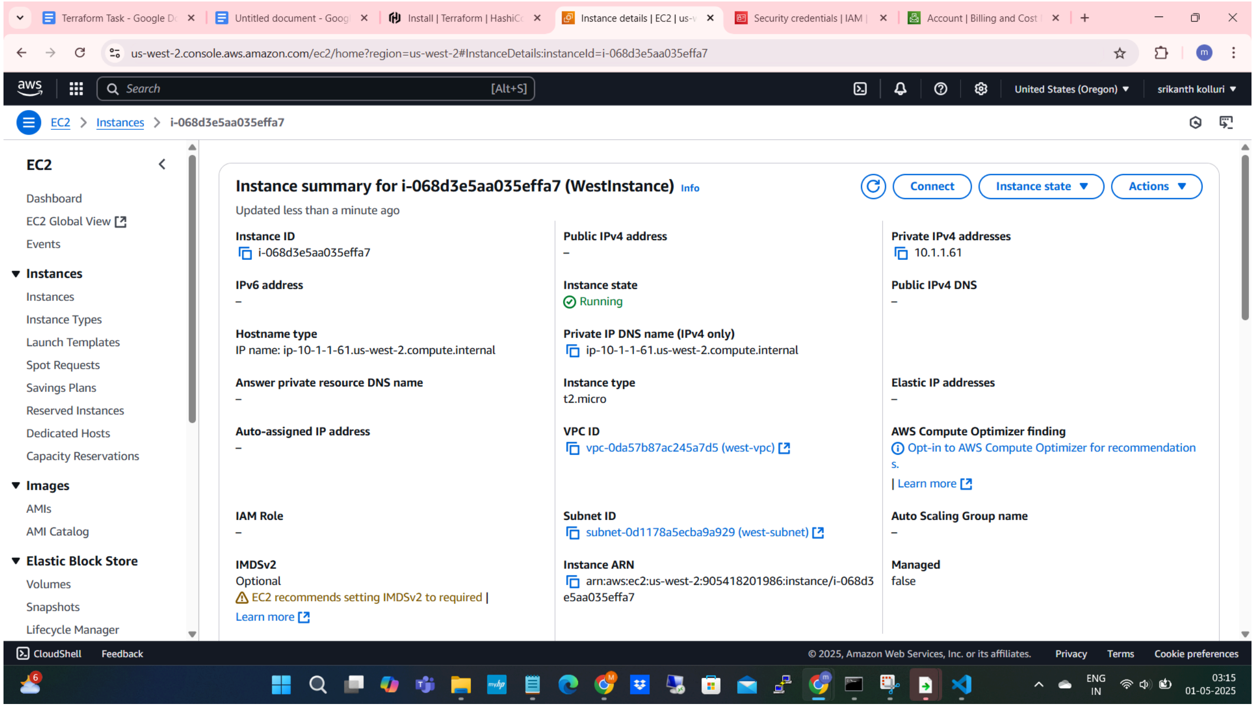
Instance created in : us-east-1 region





Instance created in : us-west-2 region





Both EC2 instances created in us-east-1 and us-east-2 regions with the single terraform